

AUDIT SURFACE FOR

Capability Matters: A Casebook

Validation & Audit

INDEXES · PER-CASE REFERENCES

EDITED BY

William Gray-Roncal, PhD
James Diamond, PhD

Learning Engineering for Next-Generation Systems · v2.1

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SECTION

What this document is for

Validation and Audit is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the LEOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

- I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.
- II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.
- III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

On verification. The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganises the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset’s standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE

Case 5

Operation Eagle Claw (1980). Cross-organizational capability that existed on no service’s deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.

STANDOUT SUCCESS

Case 18

U.S. Nuclear Navy / Rickover Training Model (1954–). The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

124	USS Fitzgerald & USS John S. McCain	2017	T·K·N
126	F-35 Sustainment & Maintainer Shortage	ongoing	T·K·D
127	Military Fratricide — Desert Storm to Afghanistan	1991 – 2014	T·H·K
128	Operation Eagle Claw	1980	T·K
129	Patriot Missile / Dhahran	1991	D·H·K
130	USS Vincennes & Iran Air Flight 655	1988	H·T
131	Stanislav Petrov / 1983 False Alert	1983	H·T
133	Mark 14 Torpedo Failures	1941 – 1943	T·K·G
134	V-22 Osprey	1991 – present	T·H·N
135	9/11 Intelligence Sharing Failures	1996 – 2001	G·K
137	U.S. Nuclear Navy / Rickover Training Model	1954 – present	T·K·N
138	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard	2020 revision (series since 1968)	G·K

139	GIFT and the Adoption Gap	2012 – present	K·G·N
140	Navy Surface Warfare Readiness Reform	2018 – present	T·K·N
141	DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks	2009 – 2014	H·K·D

AVIATION

STANDOUT FAILURE

Case 7

Boeing 737 MAX / MCAS (2018–19). A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS

Case 70

Crew Resource Management & CAST (1981–). First-officer authority engineered into the cockpit, a continuous learning architecture across the industry, and the safety record that followed.

96	AeroPerú Flight 603	1996	H·T
97	Boeing 737 MAX / MCAS	2018 – 2019	D·T·H
102	Air France Flight 447	2009	T·H
103	Kegworth / British Midland 92	1989	T·H·K
104	Asiana Airlines Flight 214	2013	T·H
105	Helios Airways Flight 522	2005	T·H
106	TransAsia Airways Flight 235	2015	T·H
107	Eastern Air Lines Flight 401	1972	H·T
109	Colgan Air Flight 3407	2009	T
110	Atlas Air Flight 3591	2019	T·K
112	NASA Saturn V Documentation	1972 – present	K
117	Crew Resource Management & CAST	1981 – present	T·H·N
118	Korean Air Safety Transformation	2000 – present	T·N
119	Aviation Safety Reporting System (ASRS)	1976 – present	T·K·N
120	EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation	1996 – 2002	H·K·G
121	TCAS — Coordinated Collision Avoidance and the Überlingen Lesson (Version 7.1)	1981 – 2008 (TCAS II)	H·K·G
122	Singapore Airlines Safety Transformation	1980s – present	T·N
132	F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap	2008 – 2012	H·K·N

HEALTHCARE

STANDOUT
FAILURE

Case 61

EHR / CPOE Implementation (2005–). Roughly \$30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT
SUCCESS

Case 109

WHO Surgical Safety Checklist (2008–). One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

1	Therac-25	1985 – 1987	H·D·G
2	EHR / CPOE Implementation	2005 – present	H·D·G
4	Mid Staffordshire NHS Foundation Trust	2005 – 2009	G·N·K
5	Epic Sepsis Model	2017 – 2021	D·G·N
8	Medical Errors as Systemic Failure	1999 – present	T·H·N·K·G
9	Vioxx Withdrawal	1999 – 2004	G·D
11	California Nurse Staffing Ratios — Legislating a Capability Requirement	2004 – 2010	G·K
12	ACGME 80-Hour Resident Duty-Hour Reform	2003–2017	T·K·N
13	Implementation Science in Healthcare — The 17-Year Gap	ongoing	K·G·N
19	Keystone ICU / Pronovost Checklist	2004 – present	T·N
20	TREWS / Sepsis Watch	2018 – 2022	H·K·G
21	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff	2024 – 2025	H·K·N
22	ChatGPT in Healthcare — Hallucination Cases	2023 – present	H·D
23	WHO Surgical Safety Checklist	2008 – present	T·N
24	Bristol Heart Babies Reform	1984 – present	G·N
32	Annual-Screening UI Redesign + CDS at University of Missouri Health Care	2020	T·D·N
33	Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal	2019 – 2024	T·D·N
34	COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case	2022 – 2024	T·K·D
35	Radiology AI Miscalibration	2018 – present	H·K·D
36	AlphaFold — Protein Structure Prediction	2020 – present	T
37	DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback	2020	T·K·D

39 TeamSTEPPS 2006 – present

T-N

EDUCATION AT SCALE

STANDOUT
FAILURE

Case 149

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT
SUCCESS

Case 110

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

45 Tennessee Voluntary Pre-K Study 2009–2018

G-D

46 Algorithmic Bias in Educational Predictive Analytics ongoing

G-H-D

51 Atlanta Public Schools Cheating Scandal 2009 – 2015

G-N

53 inBloom 2014

G

54 Summit Learning / Personalized Learning Rollout 2014–2019

G-T-K

67 Cognitive Tutor / Carnegie Learning 1990s – present

T

80 Georgia State University Predictive Analytics 2012 – present

T-K

142 xAPI / Total Learning Architecture — Interoperability Gap ongoing

K-G

205 The Discipline We Build Next ongoing

T-K-N-G-H-D

ENERGY

STANDOUT
FAILURE

Case 69

Three Mile Island (1979). Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT
SUCCESS

Case 172

INPO and the Nuclear Academy (1979–). The industry built the institution TMI revealed it needed. No INES-level event at U.S. commercial reactors since.

146 Northeast Blackout 2003

H-K

GOVERNMENT

STANDOUT FAILURE

Case 162

Australia Robodebt (2016–20). An income-averaging algorithm that reversed the burden of proof onto welfare recipients. The Royal Commission found “venality, incompetence and cowardice”; the scheme was linked to deaths, including suicide.

NOTE

Case 111

No paired-intervention success in the government dataset. The reforms that follow these cases — Royal Commissions, statutory restructures — are documented in the narratives but did not produce a LENS-style success case in time for this volume.

7 VA Wait-Time Scandal 2014

G-K-N

INDUSTRIAL

STANDOUT FAILURE

Case 147

Bhopal (1984). Training, knowledge, normalisation of deviance, and governance all collapsed in the same plant. Roughly 4,000 immediate deaths; chronic harms continue. The cautionary case for outsourced operational responsibility.

STANDOUT SUCCESS

Case 73

Toyota Production System / Andon Cord (1950s–). Any operator can stop the line. The smallest authority-architecture intervention in the dataset, and the most durably copied.

144 Kodak Digital Camera Stagnation 1975–2012

D-K-N

147 Takata Airbag Inflators 2008 – 2023

D-G

149 Volkswagen Dieselgate 2015

D-G

151 GM Ignition Switch 2002 – 2014

D-G

155 Toyota Production System / Andon Cord 1950s – present

N-G

TECHNOLOGY

STANDOUT FAILURE

Case 66

UK Post Office Horizon Scandal (1999–2015). Institutional refusal to believe sub-postmasters’ reports of computer error, sustained over fifteen years. Hundreds of wrongful convictions; ruined lives and at least four suicides — the longest-running operator-feedback failure in the dataset.

NOTE

Case 132

The technology dataset’s success story is filed under Healthcare: AlphaFold (Case 36), the contemporary worked example of computational capability-engineering done as a discipline.

143 Knight Capital Trading Loss 2012

D-K

148	Wells Fargo Fake Accounts	2011 – 2016	G·N
152	TSB Bank IT Migration	2018	H·G
153	Equifax Data Breach	2017	G·K
157	AI-Augmented Coding Tools	2021 – present	T·H

AI IN EDUCATION

88	LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models	2025	T·K·N
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K-12 EDUCATION

48	School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism	2010s – 2022	G·K·N
60	Houston EVAAS — Value-Added Teacher Evaluation on Trial	2011–2017	G
61	Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap	1997–2023	K·T
62	Gates Intensive Partnerships — Measurement Without a Coupled Lever	2009–2018	G·K
63	LAUSD iPad Program — The Common Core Technology Project	2013–2015	D·H·G
66	Growth-Mindset National Experiment — Heterogeneous Effects	2019	D·N·K
75	Chen et al. — Rural China AI Devices and the Equity-Direction Finding	2025	T·K·D
95	PBIS Implementation Fidelity — Why the Framework Travels Only with Its Measurement Loop	1997–2024	T·G

K-12 MATHEMATICS

72	ASSISTments — National Replication and Long-Term Follow-Through	2014 – present	T·K·D
84	Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive	2007 – 2014	T·K·D

K-12 SCIENCE

74	Zhang/Scardamalia — Knowledge Building Across Three Cohorts	2009	T·K·D
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AIR TRAFFIC MANAGEMENT

116	Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change	2005	G·K·H
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ANESTHESIOLOGY

27	Anesthesia Monitoring Standards and the APSF — The Mortality Decline	1986 – present	H·K·G
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AVIATION INFRASTRUCTURE

115	FAA NextGen and the ADS-B Out Transition	2003 – 2020	G·D·K
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AVIATION SAFETY

114 FAA Aging-Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule 1988 – 2010s **G·D·K**

123 FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike 1992 – 2000s **G·K·N**

CLINICAL CARE

15 I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer 2014 **H·K·N**

29 Bar-Code Medication Administration — Cue at the Point of Care 2010 **H·K·D**

CLINICAL MEDICINE

6 Racial Bias in Pain Assessment — The False-Belief Mechanism 2016 **T·K·N**

25 Removing the Race Coefficient from eGFR 2021 **D·G·N**

CLINICAL/TRANSLATIONAL RESEARCH

40 Team Science Training for Clinical and Translational Scientists 2020 – 2022 **K·N**

CORPORATE L&D

65 Training Transfer — The Gap Between Learning and Doing 2010 **K·N**

70 High-Impact Learning System — Engineering the Environment, Not Just the Event 2001 – present **K·N**

79 The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops 1959 – present **K·N**

83 Brinkerhoff Success Case Method — Tails as the Evaluation Instrument 2005 – present **K·N**

ED-TECH

47 Algorithmic Bias in Automated Exam Proctoring 2022 **D·N·K**

EDUCATION (NORWAY)

92 Norway's National Expert Commission on Learning Analytics 2021 – 2023 **G·K·N**

EDUCATION AT SCALE

50 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction 2012 – 2024 **D·K·N**

69 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale 2016 **T·D**

GENERAL AVIATION

100 Glass-Cockpit Transition in Light Aircraft — Technology Outran Training 2010 **H·N·K**

GLOBAL HEALTH

18	PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption 2023	K·N·D
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43	Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface 2013 – 2018	H·N
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GOVERNMENT

49	UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days 2020	D·K·N
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HEALTH PROFESSIONS EDUCATION

28	Interprofessional Education and the Evidence Gap 2013 – 2015	K·N
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HEALTHCARE/ONCOLOGY

3	Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018	D·K·N
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HIGHER EDUCATION

55	Enrollment-Algorithm Yield Optimization Across U.S. Higher Education (Brookings synthesis 2021) 2010s – present	T·K·N
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57	GAO Online Program Manager Oversight Gap (GAO-22-104463) 2022	D·K·N
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58	USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC) 2010s – 2024	D·K·N
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85	OU Analyse — Predictive Learning Analytics and Teacher Use at Scale 2019 – 2023	T·K·D
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86	Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction 2024	D·K·N
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90	Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions 2025	T·K·N
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HIGHER EDUCATION (AFRICA)

89	Data Privacy and Learning Analytics on the African Continent 2022	K·G·N
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HIGHER EDUCATION (BRAZIL)

82	MMALA — A Maturity Model for Responsible Learning-Analytics Adoption (Brazil) 2024	K·N
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HIGHER EDUCATION (LATIN AMERICA)

91	LALA — Building Learning-Analytics Governance Capacity Across Latin America 2017 – 2020	K·N
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HIGHER ED (UK)

81	Open University 'Ethical Use of Student Data' and OU Analyse 2014 – 2025	G·K·N
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HIGHER-ED ANALYTICS

52 Purdue Course Signals — The Reverse-Causality Retention Claim 2012 – 2013 **D-K-N**

HOSPITAL PHARMACY

42 Australian Hospital-Pharmacy Technician Role Redesign 2016 **D-N-H**

IMPLEMENTATION SCIENCE

41 Implementation-Science Training — Stated Goals Outrunning Operational Practice 2020s **K-N**

LEARNING ANALYTICS

73 The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension 2016 – 2025 **T-K-D**

87 Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models 2021 – 2024 **D-K-N**

94 Learning Analytics on the African Continent — A Scoping Review as the Present-State Map 2022 **K-N**

MEDICAL DEVICES

26 Pulse Oximetry Across Skin Tones 1990 – 2020 – 2025 **D-G-N**

MEDICAL EDUCATION

14 Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA 2009 – 2020s **T-K**

17 Spaced Education RCTs in Medical Training 2007 – 2009 **H-K-D**

MEDICAL-DEVICE REGULATION

44 Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD 2014 – present **G-N**

MULTIMODAL LEARNING ANALYTICS

76 Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account 2023 **T-K-N**

NEUROSCIENCE/CONNECTOMICS

78 CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale 2017 – present **H-K-N**

NUCLEAR ENGINEERING

156 INL / LWRs Control-Room Modernization — Sustainment Research for an Aging Fleet 2010 – present **D-H-K**

NUCLEAR POWER

154	INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding	2014	G-K-H
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NURSING EDUCATION

16	NCSBN National Simulation Study — Licensing the 50% Substitution Rule	2014	G-K-D
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PHARMA SAFETY

38	iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms	2006 – 2011	G-D-N
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SOFTWARE ENGINEERING EDUCATION

71	Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development	2025 (preprint)	K-N
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SPACE / AEROSPACE

STANDOUT FAILURE	Case 140	Challenger & Columbia (1986 / 2003). Normalisation of deviance across two decades and two crews, captured in the Rogers and CAIB reports as a culture that accepted flying with flaws when problems were defined as normal and routine.
NOTE	Case 18	No paired-intervention success in the aerospace dataset. See Defense (Rickover, Case 137) for the safety-engineering counterpart that aerospace did not build.

98	Mars Climate Orbiter — Unit Mismatch	1999	D-K
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101	Ariane 5 Flight 501	1996	D-K-H
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108	NASA EVA-23 — Water Intrusion Inside a Spacesuit	2013	H-N-D
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111	Challenger & Columbia	1986 / 2003	N-K-G
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113	Boeing Starliner	2019 – 2024	K-D
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SURGERY

30	Surgical Skill Video Peer-Rating Predicts Complications	2013	H-D-K
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31	Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units	2009 – 2016	T-K-H
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TUTORING

77	Hybrid Human-AI Tutoring — Augmentation, Not Delegation	2024	T-K-D
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WORKFORCE DEV

68	CIRCUIT — A Scalable, Equity-Centered Research Workforce Model	2017 – 2023 (six cycles)	T-K
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93 Singapore SkillsFuture — National Workforce Capability at Scale 2015 – present

G·K·D

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the specialization, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

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25 cases

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LEN 3

37 cases

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56 cases

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Every reference cited in every case, organised by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds an explicit **Retrieved from:** line pointing to the canonical access path — a publisher URL, a DOI, an agency landing page, or a stable archival locator. Where that line is left blank, the source is under verification; the running status of that work is in casebook/verification-log.md. Future editions will fill in remaining locators as the per-case review closes them out.

Style. References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

CASE 1 Therac-251985 – 1987

1. N. G. Leveson & C. S. Turner, "An Investigation of the Therac-25 Accidents," IEEE Computer 26(7): 18–41 (1993). doi:10.1109/MC.1993.274940 — the removed hardware interlocks and the adapted software. — Retrieved from: -----

2. Leveson & Turner (1993) — the two software defects (data-entry race condition and counter overflow), the uninformative "Malfunction 54", overdoses up to 100×, six accidents, and at least three deaths. — Retrieved from: -----

3. Leveson & Turner (1993); N. G. Leveson, Safeware: System Safety and Computers (Addison-Wesley, 1995) — the manufacturer's denial and the systemic findings. — Retrieved from: -----

4. N. G. Leveson, Engineering a Safer World: Systems Thinking Applied to Safety (MIT Press, 2011) — why removing a safeguard requires explicitly reassigning its safety function. — Retrieved from: -----

5. The case's role in medical-device software regulation and software-safety practice (FDA software guidance; IEC 62304 lineage); see also Ethics Unwrapped, UT Austin. — Retrieved from: -----

CASE 2 EHR / CPOE Implementation2005 – present

1. Health Information Technology for Economic and Clinical Health (HITECH) Act (2009) — the \$30B EHR incentive program. — Retrieved from: -----

2. Y. Han et al., "Unexpected Increased Mortality After Implementation of a Commercially Sold CPOE System," Pediatrics 116(6): 1506–1512 (2005) — the disputed pediatric-ICU mortality result. — Retrieved from: -----

3. KLAS Arch Collaborative, EHR usability and safety surveys (2023) — usability as the top clinician complaint, correlated with safety outcomes. — Retrieved from: -----

4. D. Sittig & H. Singh, "Defining health information technology-related errors," and related EHR-safety work (2011–2013). — Retrieved from: -----

5. AHRQ Patient Safety Network, EHR-related harm cases — the default-result delay and the unactivated-alert overdose. — Retrieved from: _____
6. AMA / Pew / MedStar, EHR Usability and Patient Safety (2018); R. Wachter, The Digital Doctor (2015). — Retrieved from: _____

CASE 3 Watson for Oncology — Delegation Marketed Ahead of Capability

2013 – 2018

1. Ross & Swetlitz (2017–2018), “IBM Watson recommended unsafe and incorrect cancer treatments” and adjacent investigations, STAT News — the load-bearing primary source; investigative journalism drawing on leaked IBM internal documents. — Retrieved from: _____
2. Strickland (2019), “How IBM Watson Overpromised and Underdelivered on AI Health Care,” IEEE Spectrum — independent retrospective synthesis of the public record. — Retrieved from: _____
3. Topol (2019), Deep Medicine, Basic Books — secondary academic situating of Watson within the broader clinical-AI delegation discourse. — Retrieved from: _____
4. v2 paired cases: TREWS (Case 20), Epic Sepsis Model (Case 5), SyRI (Case 189) — the AI-delegation typology. — Retrieved from: _____

CASE 4 Mid Staffordshire NHS Foundation Trust

2005 – 2009

1. R. Francis QC, Report of the Mid Staffordshire NHS Foundation Trust Public Inquiry (2013) — the staffing cuts and Foundation Trust targets. — Retrieved from: _____
2. Healthcare Commission, Investigation into Mid Staffordshire NHS Foundation Trust (2009) — ward conditions and excess mortality. — Retrieved from: _____
3. Francis QC (2013) — the 2,000-page report and 290 recommendations. — Retrieved from: _____
4. Francis QC (2013) — “the system as a whole failed in its most essential duty — to protect patients from unacceptable risks of harm” (quoted). — Retrieved from: _____
5. D. Berwick, A Promise to Learn — A Commitment to Act (National Advisory Group on the Safety of Patients in England, 2013). — Retrieved from: _____
6. K. Walshe & J. Higgins (2002) on NHS safety governance; cf. the VA wait-time scandal (Case 7). — Retrieved from: _____

CASE 5 Epic Sepsis Model

2017 – 2021

1. Wong et al. (2021), “External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients,” JAMA Internal Medicine 181(8):1065–1070, doi:10.1001/jamainternmed.2021.2626. — Retrieved from: _____
2. Habib et al. (2021), commentary on Wong et al., JAMA Internal Medicine — on the implications for proprietary clinical AI. — Retrieved from: _____
3. FDA, Clinical Decision Support Software: Final Guidance (2022) — the post-Wong reframing of the EHR-embedded oversight question. — Retrieved from: _____
4. Adams et al. (2022), Nature Medicine — the paired positive case (20). — Retrieved from: _____

CASE 6 Racial Bias in Pain Assessment — The False-Belief Mechanism

2016

1. Hoffman, Trawalter, Axt, & Oliver (2016), “Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites,” PNAS 113(16):4296–4301, doi:10.1073/pnas.1516047113. — Retrieved from: _____

2. Anderson, Green, & Payne (2009), "Racial and ethnic disparities in pain: causes and consequences of unequal care," *Journal of Pain* 10(12):1187–1204 — the population-level disparity. — Retrieved from: _____
3. Sabin & Greenwald (2012), "The influence of implicit bias on treatment recommendations for 4 common pediatric conditions," *American Journal of Public Health* — the diffuse-mechanism backdrop the Hoffman precision improves on. — Retrieved from: _____
4. Vyas, Eisenstein, & Jones (2020), *NEJM* — connecting race-in-clinical-algorithms to race-in-clinical-judgment. — Retrieved from: _____

CASE 7 VA Wait-Time Scandal

2014

1. VA Office of Inspector General, Report 14-02603-267 (2014) — secret waiting lists and falsified appointment data. — Retrieved from: _____
2. GAO Veterans Health Administration reports (2000–2019) — fifteen years of data-reliability warnings. — Retrieved from: _____
3. VA OIG reports (2005, 2007, 2008) — prior, unactioned findings. — Retrieved from: _____
4. D. Draper (GAO), House VA Committee testimony, GAO-19-687T (2019) — "schedulers are among the top ten highest-turnover positions in the VA" (quoted). — Retrieved from: _____
5. Veterans Access, Choice, and Accountability Act (2014) — the legislative response. — Retrieved from: _____
6. C. Argyris & D. Schön, *Organizational Learning* (1978); measurement-gaming literature. — Retrieved from: _____

CASE 8 Medical Errors as Systemic Failure

1999 – present

1. Institute of Medicine, *To Err Is Human: Building a Safer Health System* (1999); *Crossing the Quality Chasm* (2001); *Improving Diagnosis in Health Care* (2015) — the field-defining trilogy and the 44,000–98,000 estimate; the systems framing. — Retrieved from: _____
2. Makary, M. & Daniel, M. (2016), "Medical error — the third leading cause of death in the US," *BMJ* 353:i2139 — the 250,000 estimate, the quoted framing, the extrapolation from four prior studies (IOM 1999; OIG 2010; Landrigan 2010; Classen 2011). — Retrieved from: _____
3. Shojania, K. & Dixon-Woods, M. (2017), "Estimating deaths due to medical error: the ongoing controversy and why it matters," *BMJ Quality & Safety* 26(5):423–428; with companion commentaries including Carr in *Health Affairs* — methodological contestation of the Makary extrapolation and CDC-ranking comparison. — Retrieved from: _____
4. Makary & Daniel (2016), *BMJ* — death certificates do not capture medical error as a cause; ICD billing taxonomy as the structural reason. — Retrieved from: _____
5. Bates, D. W., Levine, D. M., Salmasian, H., et al. (2023), "The Safety of Inpatient Health Care," *NEJM* 388(2):142–153 — eleven-hospital Massachusetts cohort; adverse events in 25% of admissions, 25% of those preventable; persistence of harm at scale. — Retrieved from: _____
6. Agency for Healthcare Research and Quality, *National Healthcare Quality and Disparities Reports* (annual); CDC WONDER ICD-coded mortality data — institutional context for the missing national active-surveillance instrument. — Retrieved from: _____

CASE 9 Vioxx Withdrawal

1999 – 2004

1. Bombardier, C. et al. (2000), "Comparison of upper gastrointestinal toxicity of rofecoxib and naproxen in patients with rheumatoid arthritis," *VIGOR trial, NEJM* 343(21):1520–1528 — the five-fold myocardial-infarction signal and the naproxen-protective hypothesis. — Retrieved from: _____

2. Bresalier, R. et al. (2005), "Cardiovascular events associated with rofecoxib in a colorectal adenoma chemoprevention trial," APPROVe, NEJM 352(11):1092–1102 — placebo-controlled confirmation of cardiovascular risk; early trial termination. — Retrieved from: _____
3. Graham, D. J. et al. (2005), "Risk of acute myocardial infarction and sudden cardiac death in patients treated with COX-2 selective and non-selective NSAIDs: nested case-control study," Lancet 365(9458):475–481 — Kaiser Permanente population-level cardiovascular risk analysis. — Retrieved from: _____
4. US Senate Committee on Finance, hearings on Vioxx and FDA's drug-safety system (November 18, 2004) — Graham testimony; "88,000 to 139,000 Americans" estimate of excess cardiovascular events; described FDA management pressure. — Retrieved from: _____
5. Ross, J. S., Hill, K. P., Egilman, D. S., Krumholz, H. M. (2008), "Guest authorship and ghostwriting in publications related to rofecoxib," JAMA 299(15):1800–1812 — publication-bias and ghost-authorship documentation from Merck litigation discovery. — Retrieved from: _____
6. FDA Amendments Act of 2007 (P.L. 110-85) and FDA Sentinel Initiative documentation (2008–present) — REMS authority, post-market study requirements, and active distributed-data post-market surveillance. — Retrieved from: _____

CASE 11 **California Nurse Staffing Ratios — Legislating a Capability Requirement** 2004 – 2010

1. Aiken, Sloane, Cimiotti, Clarke, Flynn, Seago, Spetz, & Smith (2010), "Implications of the California Nurse Staffing Mandate for Other States," Health Services Research 45(4):904–921, doi:10.1111/j.1475-6773.2010.01114.x. — Retrieved from: _____
2. Aiken, Clarke, Sloane, Sochalski, & Silber (2002), "Hospital nurse staffing and patient mortality, nurse burnout, and job dissatisfaction," JAMA 288(16):1987–1993 — the workload-mortality relationship the 2010 modeling rests on. — Retrieved from: _____
3. California Department of Health Services (1999–2004), AB 394 regulatory implementation documentation. — Retrieved from: _____
4. Spetz, Chapman, Herrera, Kaiser, Seago, & Dower (2009), "Assessing the impact of California's nurse staffing ratios on hospitals and patient care," California HealthCare Foundation — implementation-period analysis. — Retrieved from: _____

CASE 12 **ACGME 80-Hour Resident Duty-Hour Reform** 2003–2017

1. B. H. Lerner, The Libby Zion Case and the Reform of Medical Education (2006); and the 1989 New York State (Bell Commission) duty-hour regulations — the origin of the reform. — Retrieved from: _____
2. Accreditation Council for Graduate Medical Education, Common Program Requirements (2003 and 2011 revisions) — the 80-hour weekly cap and the 16-hour first-year shift limit. — Retrieved from: _____
3. D. A. Asch et al., "Resident Duty Hours and Medical Education Policy," NEJM 370: 1671–1673 (2014); Institute of Medicine (Ulmer, Wolman & Johns, eds.), Resident Duty Hours: Enhancing Sleep, Supervision, and Safety (2008) — hand-offs and continuity trade-offs. — Retrieved from: _____
4. Bilimoria, Chung, Hedges et al., "National Cluster-Randomized Trial of Duty-Hour Flexibility in Surgical Training" (FIRST), NEJM 374: 713–727 (2016). doi:10.1056/NEJMoa1515724. — Retrieved from: _____
5. Silber, Bellini, Shea et al., "Patient Safety Outcomes under Flexible and Standard Resident Duty-Hour Rules" (iCOMPARE), NEJM 380: 905–914 (2019). doi:10.1056/NEJMoa1810641. — Retrieved from: _____
6. ACGME, Common Program Requirements (2017 revision) — relaxation of the 16-hour first-year shift limit. — Retrieved from: _____

7. P. Pronovost et al., Keystone ICU intervention, NEJM 355: 2725–2732 (2006), doi:10.1056/NEJMoa061115 A. B. Haynes et al., surgical safety checklist, NEJM 360: 491–499 (2009), doi:10.1056/NEJMsa0810119 — integrated interventions that engineered the surrounding architecture. — Retrieved from: _____

CASE 13 Implementation Science in Healthcare — The 17-Year Gap

ongoing

1. E. A. Balas & S. A. Boren (2000), Yearbook of Medical Informatics — the 17-year / 14% translation figures. — Retrieved from: _____
2. Z. Morris, S. Wooding & J. Grant, “The answer is 17 years, what is the question,” J. Royal Society of Medicine (2011) (quoted). — Retrieved from: _____
3. D. Fixsen et al., Implementation Research: A Synthesis of the Literature (2005) — the Active Implementation Frameworks. — Retrieved from: _____
4. G. Aarons et al. (2011), the EPIS framework; L. Damschroder et al. (2009), CFIR. — Retrieved from: _____
5. Cf. medical error as systemic failure (Case 8); Goodell & Kolodner, Learning Engineering Toolkit (2022). — Retrieved from: _____

CASE 14 Barsuk SBML — Simulation-Based Mastery Learning Dissemination from Northwestern to the VA

2009 – 2020s

1. Barsuk, Cohen, Feinglass, McGaghie, & Wayne (2009), “Use of simulation-based education to reduce catheter-related bloodstream infections,” Archives of Internal Medicine 169(15):1420–1423, doi:10.1001/archinternmed.2009.215. — Retrieved from: _____
2. Cohen, Feinglass, Barsuk, Barnard, O'Donnell, McGaghie, & Wayne (2010), “Cost savings from reduced catheter-related bloodstream infection after simulation-based education for residents in a medical intensive care unit,” Simulation in Healthcare 5(2):98–102, doi:10.1097/SIH.0b013e3181bc8304. — Retrieved from: _____
3. Barsuk, McGaghie, Cohen, Balachandran, & Wayne (2009), “Use of simulation-based mastery learning to improve the quality of central venous catheter placement in a medical intensive care unit,” Journal of Hospital Medicine 4(7):397–403. — Retrieved from: _____
4. McGaghie, Issenberg, Cohen, Barsuk, & Wayne (2011), “Does simulation-based medical education with deliberate practice yield better results than traditional clinical education? A meta-analytic comparative review,” Academic Medicine 86(6):706–711. — Retrieved from: _____
5. Department of Veterans Affairs SimLEARN documentation — the operating-program record of the national dissemination effort. — Retrieved from: _____

CASE 15 I-PASS Handoff Bundle — Structuring the Human-to-Human Transfer

2014

1. Starmer, A. J., Spector, N. D., Srivastava, R., West, D. C., Rosenbluth, G., Allen, A. D., Noble, E. L., Tse, L. L., Dalal, A. K., Keohane, C. A., Lipsitz, S. R., Rothschild, J. M., Wien, M. F., Yoon, C. S., Zigmont, K. R., Wilson, K. M., O'Toole, J. K., Solan, L. G., Aylor, M., Bismilla, Z., Coffey, M., Mahant, S., Blankenburg, R. L., Destino, L. A., Everhart, J. L., Patel, S. J., Bale, J. F., Spackman, J. B., Stevenson, A. T., Calaman, S., Cole, F. S., Balmer, D. F., Hepps, J. H., Lopreiato, J. O., Yu, C. E., Sectish, T. C., & Landrigan, C. P. (2014). Changes in medical errors after implementation of a handoff program. New England Journal of Medicine, 371(19):1803–1812. doi:10.1056/NEJMsa1405556 — the case's primary evaluation. — Retrieved from: _____
2. Starmer, A. J., et al. (2014). Implementation correspondence and follow-up. New England Journal of Medicine — the published correspondence cautioning against implementing the mnemonic alone. — Retrieved from: _____
3. Sectish, T. C., et al. (2010). Establishing a multisite education and research project: The I-PASS Study Group. Academic Medicine — the I-PASS Study Group methodology and design rationale. — Retrieved from: _____

4. Cohen, M. D., & Hilligoss, P. B. (2010). The published literature on handoffs in hospitals: deficiencies identified in an extensive review. *Quality and Safety in Health Care — the broader handoff-failure-mode literature.* — Retrieved from: _____

CASE 16 NCSBN National Simulation Study — Licensing the 50% Substitution Rule 2014

1. Hayden, J. K., Smiley, R. A., Alexander, M., Kardong-Edgren, S., & Jeffries, P. R. (2014). The NCSBN National Simulation Study: A longitudinal, randomized, controlled study replacing clinical hours with simulation in prelicensure nursing education. *Journal of Nursing Regulation*, 5(2 Suppl):S1–S64. [https://www.journalofnursingregulation.com/article/s2155-8256\(15\)30062-4/fulltext](https://www.journalofnursingregulation.com/article/s2155-8256(15)30062-4/fulltext) — the case's primary study. — Retrieved from: _____
2. Smiley, R. A., & Martin, B. (2023). Simulation in Nursing Education: Advancements in Regulation, 2014–2022. *Journal of Nursing Regulation*. doi:10.1016/S2155-8256(23)00086-8 — the 2023 follow-up documenting the regulatory propagation. — Retrieved from: _____
3. Jeffries, P. R. (2012). *Simulation in Nursing Education: From Conceptualization to Evaluation* (2nd ed.). National League for Nursing — the simulation-quality framework the NCSBN study's quality conditions rest on. — Retrieved from: _____
4. INACSL Standards Committee (2016). INACSL Standards of Best Practice: Simulation. *Clinical Simulation in Nursing*, 12(S) — the simulation-practice standards downstream programs adopt as the analog of the NCSBN quality conditions. — Retrieved from: _____

CASE 17 Spaced Education RCTs in Medical Training 2007 – 2009

1. Kerfoot, B. P., Baker, H. E., Koch, M. O., Connelly, D., Joseph, D. B., & Ritchey, M. L. (2007). Randomized, controlled trial of spaced education to urology residents in the United States and Canada. *Journal of Urology*, 177(4):1481–1487. doi:10.1016/j.juro.2006.11.074 — the case's primary RCT. — Retrieved from: _____
2. Kerfoot, B. P. (2009). Learning benefits of on-line spaced education persist for 2 years. *Journal of Urology*, 181(6):2671 — the two-year persistence follow-up. — Retrieved from: _____
3. Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. *Psychological Bulletin*, 132(3):354–380 — the basic-literature spacing-effect review the workplace-L&D RCT sits against. — Retrieved from: _____
4. Settles, B., & Meeder, B. (2016). A trainable spaced repetition model for language learning. *Proceedings of ACL 2016*, 1848–1858 — Duolingo half-life regression (Case 69), the operational deployment companion. — Retrieved from: _____

CASE 18 PEPFAR HIV Training Across 16 Sub-Saharan African Countries — Modality Comparison Under Disruption 2023

1. "Comparing in-person, blended and virtual training interventions; a real-world evaluation of HIV capacity building programs in 16 countries in sub-Saharan Africa," medRxiv 2023.02.08.23285641 (preprint) → PMC10365303 (published). — Retrieved from: _____
2. PEPFAR program documentation — workforce-capability training as a load-bearing deliverable across Sub-Saharan African deployment countries. — Retrieved from: _____
3. Kirkpatrick & Kirkpatrick (2006), *Evaluating Training Programs* — the chain-of-evidence framework the L1–L2 limitation references (paired Case 79). — Retrieved from: _____
4. Blume et al. (2010), *Journal of Management* 36(4):1065–1105 — the meta-analytic transfer finding the L3 question references (paired Case 65). — Retrieved from: _____

CASE 19 Keystone ICU / Pronovost Checklist 2004 – present

1. Pronovost, P. et al. (2006), "An Intervention to Decrease Catheter-Related Bloodstream Infections in the ICU," *NEJM* 355 — the trial and the near-zero result. — Retrieved from: _____

2. Pronovost & Vohr (2010), Safe Patients, Smart Hospitals — the checklist-plus-nurse-authority pairing (paraphrased). — Retrieved from: _____

3. Lipitz-Snyderman, A. et al. (2011), BMJ — sustained effect at follow-up. — Retrieved from: _____

4. Agency for Healthcare Research and Quality, CUSP toolkit — dissemination across states. — Retrieved from: _____

5. Bosk, C. et al. (2009), “Reality check for checklists,” The Lancet — the authorization, not the list, as the active ingredient. — Retrieved from: _____

CASE 20 TREWS / Sepsis Watch

2018 – 2022

1. Adams et al. (2022), “Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis,” Nature Medicine 28(7):1455–1460, doi:10.1038/s41591-022-01894-0. — Retrieved from: _____

2. Henry et al. (2022), “Factors driving provider adoption of the TREWS machine learning-based early warning system and its effects on sepsis treatment timing,” Nature Medicine 28(7):1447–1454, doi:10.1038/s41591-022-01895-z. — Retrieved from: _____

3. Sendak et al. (2020), “Real-world integration of a sepsis deep learning technology into routine clinical care: implementation study,” JMIR Medical Informatics 8(7):e15182 (Sepsis Watch implementation). — Retrieved from: _____

4. Wong et al. (2021), “External Validation of a Widely Implemented Proprietary Sepsis Prediction Model,” JAMA Internal Medicine 181(8):1065–1070 — the foil case (102). — Retrieved from: _____

CASE 21 Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff

2024 – 2025

1. Treloar, E., Herath, M., Altree, M., Potter, S., Penhall, M., Walsh, D., Kennedy, L., Bruening, M., Edwards, S., Ey, J. D., Bradshaw, E. L., & Maddern, G. J. (2025), “A Simple Solution for a Complex Problem: The ‘Sterile Cockpit’ to Improve Ward Rounds,” World Journal of Surgery 49(10):2769–2776, doi:10.1002/wjs.70074, PMID:40930848, PMCID:PMC12515027 — the cited adaptation study. — Retrieved from: _____

2. Federal Aviation Administration, 14 CFR § 121.542 (codified 1981) — origin of the aviation sterile-cockpit rule. — Retrieved from: _____

3. Starmer, A. J. et al. (2014), “Changes in medical errors after implementation of a handoff program,” New England Journal of Medicine 371(19):1803–1812 — I-PASS handoff intervention; structural cousin in the structured-information half of handoff capability. — Retrieved from: _____

4. Broom, M. A., Capek, A. L., Carachi, P., Akeroyd, M. A., & Hilditch, G. (2011), “Critical phase distractions in anaesthesia and the sterile cockpit concept,” Anaesthesia 66(3):175–179 — prior anaesthesia-domain sterile-cockpit adaptation establishing the cross-domain transfer pattern. — Retrieved from: _____

CASE 22 ChatGPT in Healthcare — Hallucination Cases

2023 – present

1. JAMA editorials on LLM integration into clinical practice (2023–2024) — the hallucination/verification problem (synthesized). — Retrieved from: _____

2. Sallam (2023), “ChatGPT Utility in Healthcare Education, Research, and Practice” — documented benefits and risks. — Retrieved from: _____

3. Case reports of LLM hallucinated citations and dosages in clinical and academic use (2023–2024). — Retrieved from: _____

4. WHO ethical guidance on AI for health (2024) — verification and oversight requirements. — Retrieved from: _____

5. Wachter & Brynjolfsson (2023), JAMA — generative AI in health care. — Retrieved from: _____

CASE 23 WHO Surgical Safety Checklist

2008 – present

1. Haynes, A. et al. (2009), “A Surgical Safety Checklist to Reduce Morbidity and Mortality,” NEJM — the 1.5%→0.8% result and major-complication reduction. — Retrieved from: _____
2. WHO Safe Surgery Saves Lives campaign documentation — the nineteen-item checklist and the three junctures. — Retrieved from: _____
3. Gawande, A. (2009), The Checklist Manifesto — the pause as the active mechanism. — Retrieved from: _____
4. Urbach, D. et al. (2014), “Introduction of Surgical Safety Checklists in Ontario, Canada,” NEJM — null mortality result after mandate. — Retrieved from: _____
5. Bosk, C. et al. (2009), “Reality check for checklists,” The Lancet — implementation fidelity. — Retrieved from: _____

CASE 24 Bristol Heart Babies Reform

1984 – present

1. Kennedy, I. (2001), Learning from Bristol: The Report of the Public Inquiry into Children’s Heart Surgery at the Bristol Royal Infirmary 1984–1995 — the inquiry and recommendations (paraphrased). — Retrieved from: _____
2. Society for Cardiothoracic Surgery in GB & Ireland, national outcomes reports — the registry and surgeon-level publication. — Retrieved from: _____
3. Berwick, D. (2013), A Promise to Learn — A Commitment to Act — the broader NHS–safety reform. — Retrieved from: _____
4. Sherlaw-Johnson et al. — risk-adjusted outcome methodology. — Retrieved from: _____
5. NHS national clinical audit and registry documentation — extension across specialties. — Retrieved from: _____

CASE 25 Removing the Race Coefficient from eGFR

2021

1. Delgado et al. (2021), “A Unifying Approach for GFR Estimation: Recommendations of the NKF–ASN Task Force on Reassessing the Inclusion of Race in Diagnosing Kidney Disease,” American Journal of Kidney Diseases 79(2):268–288 (published online 2021; in print vol. 79, 2022), doi:10.1053/j.ajkd.2021.08.003. Cited by online–first year, the year of the Task Force recommendation. — Retrieved from: _____
2. Inker et al. (2021), “New Creatinine- and Cystatin C–Based Equations to Estimate GFR without Race,” New England Journal of Medicine 385(19):1737–1749, doi:10.1056/NEJMoa2102953. — Retrieved from: _____
3. Eneanya, Yang, & Reese (2019), “Reconsidering the Consequences of Using Race to Estimate Kidney Function,” JAMA 322(2):113–114 — the equity argument that motivated the revision. — Retrieved from: _____
4. Vyas, Eisenstein, & Jones (2020), “Hidden in Plain Sight — Reconsidering the Use of Race Correction in Clinical Algorithms,” NEJM 383(9):874–882 — broader race-in-algorithms survey. — Retrieved from: _____

CASE 26 Pulse Oximetry Across Skin Tones

1990 – 2020 – 2025

1. Sjöding, Dickson, Iwashyna, Gay, & Valley (2020), “Racial Bias in Pulse Oximetry Measurement,” New England Journal of Medicine 383(25):2477–2478, doi:10.1056/NEJMc2029240. — Retrieved from: _____
2. Jubran & Tobin (1990), “Reliability of Pulse Oximetry in Titrating Supplemental Oxygen Therapy in Ventilator-Dependent Patients,” Chest 97(6):1420–1425 — original finding, published thirty years earlier. — Retrieved from: _____

3. FDA (2025), "Pulse Oximeters for Medical Purposes — Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations: Draft Guidance for Industry and Food and Drug Administration Staff," issued January 7 2025, Docket No. FDA-2023-N-4976; Federal Register notice 2024-31540 — the regulatory corrective-action artifact, language may evolve in final. — Retrieved from: _____
4. Fawzy et al. (2022), "Racial and Ethnic Discrepancy in Pulse Oximetry and Delayed Identification of Treatment Eligibility Among Patients With COVID-19," JAMA Internal Medicine — downstream effect during the pandemic. — Retrieved from: _____

CASE 27 **Anesthesia Monitoring Standards and the APSF — The Mortality Decline** 1986 – present

1. Eichhorn, Cooper, Cullen, Maier, Philip, & Seeman (1986), "Standards for Patient Monitoring During Anesthesia at Harvard Medical School," JAMA 256(8):1017–1020. — Retrieved from: _____
2. Eichhorn (1989), "Prevention of intraoperative anesthesia accidents and related severe injury through safety monitoring," Anesthesiology 70(4):572–577. — Retrieved from: _____
3. Anesthesia Patient Safety Foundation (1985 – present), founding documents and the APSF Newsletter — institutional-history record of the broader change effort. — Retrieved from: _____
4. American Society of Anesthesiologists (1986), "Standards for Basic Anesthetic Monitoring" — original ASA standard. — Retrieved from: _____
5. Sjoding et al. (2020), NEJM — the racial-bias edge case the standard still carries (cross-reference Case 26). — Retrieved from: _____

CASE 28 **Interprofessional Education and the Evidence Gap** 2013 – 2015

1. Reeves, Perrier, Goldman, Freeth, & Zwarenstein (2013), "Interprofessional education: effects on professional practice and healthcare outcomes (update)," Cochrane Database of Systematic Reviews, doi:10.1002/14651858.CD002213.pub3. — Retrieved from: _____
2. Institute of Medicine (2015), Measuring the Impact of Interprofessional Education on Collaborative Practice and Patient Outcomes, National Academies Press, NCBI NBK338352. — Retrieved from: _____
3. WHO (2010), Framework for Action on Interprofessional Education and Collaborative Practice — the international policy backdrop. — Retrieved from: _____
4. v2 paired cases: Team-science training (121), Implementation-science training (123). — Retrieved from: _____

CASE 29 **Bar-Code Medication Administration — Cue at the Point of Care** 2010

1. Poon, E. G., Keohane, C. A., Yoon, C. S., Dittmore, M., Bane, A., Levitzion-Korach, O., Moniz, T., Rothschild, J. M., Kachalia, A. B., Hayes, J., Churchill, W. W., Lipsitz, S., Whittemore, A. D., Bates, D. W., & Gandhi, T. K. (2010). Effect of bar-code technology on the safety of medication administration. New England Journal of Medicine, 362(18):1698–1707. doi:10.1056/NEJMs0907115 — the case's primary evaluation. — Retrieved from: _____
2. Bonkowski, J., Carnes, C., Melucci, J., Mirtallo, J., Prier, B., Reichert, E., Moffatt-Bruce, S., & Weber, R. J. (2013). Effect of barcode-assisted medication administration on emergency department medication errors. Academic Emergency Medicine, 20(8):801–806 — adjacent transfer evidence. — Retrieved from: _____
3. Thompson, K. M., Swanson, K. M., Cox, D. L., Kirchner, R. B., Russell, J. J., Wermers, R. A., Storlie, C. B., Johnson, M. G., & Naessens, J. M. (2018), "Implementation of Bar-Code Medication Administration to Reduce Patient Harm," Mayo Clinic Proceedings: Innovations, Quality & Outcomes 2(4):342–351, doi:10.1016/j.mayocpiqo.2018.09.001, PMID:30560236, PMCID:PMC6257885 — later single-site rollout reporting a 55.4% reduction in actual patient-harm events. — Retrieved from: _____

4. Institute for Safe Medication Practices, Guidelines for Safe Electronic Communication of Medication Information — the institutional-commitment literature the workflow change rests on. — Retrieved from: _____

CASE 30 Surgical Skill Video Peer-Rating Predicts Complications

2013

1. Birkmeyer, J. D., Finks, J. F., O'Reilly, A., Oerline, M., Carlin, A. M., Nunn, A. R., Dimick, J., Banerjee, M., & Birkmeyer, N. J. O., for the Michigan Bariatric Surgery Collaborative. (2013). Surgical skill and complication rates after bariatric surgery. *New England Journal of Medicine*, 369(15):1434–1442. doi:10.1056/NEJMsa1300625 — the case's primary evaluation. — Retrieved from: _____
2. Birkmeyer, J. D., Stukel, T. A., Siewers, A. E., Goodney, P. P., Wennberg, D. E., & Lucas, F. L. (2003). Surgeon volume and operative mortality in the United States. *New England Journal of Medicine*, 349(22):2117–2127 — the volume-outcome literature the confound rests against. — Retrieved from: _____
3. Vassiliou, M. C., et al. (2005). The MISTELS program — global objective assessment of laparoscopic skills. *Surgical Endoscopy* — the surgical-skill assessment literature the rating scale derives from. — Retrieved from: _____
4. Jacobs, D. O. (2013). Cut well, sew well, do well? *New England Journal of Medicine*, 369(15):1466–1467. doi:10.1056/NEJMe1309785 — the accompanying editorial. — Retrieved from: _____

CASE 31 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units

2009 – 2016

1. Vedula, Malpani, Tao, Chen, Gao, Poddar, Ahmidi, Paxton, Vidal, Khudanpur, Hager, & Chen (2016), "Analysis of the Structure of Surgical Activity for a Suturing and Knot-Tying Task," *PLOS One* 11(3):e0149174, doi:10.1371/journal.pone.0149174. — Retrieved from: _____
2. Gao, Vedula, Reiley, Ahmidi, Varadarajan, Lin, Tao, Zappella, Bejar, Yuh, Chen, Vidal, Khudanpur, & Hager (2014), "JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS): A Surgical Activity Dataset for Human Motion Modeling," *MICCAI Workshop* — JIGSAWS dataset release paper. — Retrieved from: _____
3. Reiley, Lin, Yuh, & Hager (2011), "Review of methods for objective surgical skill evaluation," *Surgical Endoscopy* 25(2):356–366 — situates the decomposition within the broader skill-assessment literature. — Retrieved from: _____
4. Birkmeyer, Finks, O'Reilly, et al. (2013), *NEJM* — the paired skill-matters study (Case 30). — Retrieved from: _____

CASE 32 Annual-Screening UI Redesign + CDS at University of Missouri Health Care

2020

1. Pierce RP, Eskridge BR, Rehard L, Ross B, Day MA, Belden JL (2020), "The Effect of Electronic Health Record Usability Redesign on Annual Screening Rates in an Ambulatory Setting," *Applied Clinical Informatics* 11(4):580–588, doi:10.1055/s-0040-1715828. HIMSS (Greater Kansas City chapter) case study is the secondary write-up. — Retrieved from: _____
2. Co et al. (2019), "Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review," *JAMIA* 26(10):1141, doi:10.1093/jamia/ocz095 — adjacent systematic-review evidence on interaction-design redesign. — Retrieved from: _____
3. Office of the National Coordinator for Health IT, SAFER Guides on CDS design — practitioner-tier guidance the redesign instantiates. — Retrieved from: _____
4. Middleton et al. (2013), "Enhancing patient safety and quality of care by improving the usability of electronic health record systems," *JAMIA* 20(e1):e2–e8 — the framing peer-reviewed paper on EHR-usability-as-safety. — Retrieved from: _____

CASE 33

Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal

2019 – 2024

1. Hussain et al. (2019), “Medication safety alert fatigue may be reduced via interaction design and clinical role tailoring: a systematic review,” JAMIA 26(10):1141–1149, doi:10.1093/jamia/ocz095. — Retrieved from: _____
2. Patterson E. (2024), “Navigating Alert Fatigue: A Case Study in Electronic Health Record Alert Design Optimization,” Studies in Health Technology and Informatics, PubMed 39049299 — nursing-workflow QI redesign of four high-firing alerts. — Retrieved from: _____
3. Authors (2024), “Addressing Alert Fatigue by Replacing a Burdensome Interruptive Alert with Passive Clinical Decision Support,” Applied Clinical Informatics — interruptive-to-passive conversion with dual outcomes. — Retrieved from: _____
4. Office of the National Coordinator for Health IT, SAFER Guides on CDS — practitioner-tier guidance the redesigns instantiate. — Retrieved from: _____
5. Ancker et al. (2017), “Effects of workload, work complexity, and repeated alerts on alert fatigue in a clinical decision support system,” BMC Medical Informatics and Decision Making 17:36 — adjacent measurement evidence. — Retrieved from: _____

CASE 34

COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case

2022 – 2024

1. Boussina, A., Shashikumar, S. P., Malhotra, A., Owens, R. L., El-Kareh, R., Longhurst, C. A., Quintero, K., et al. (2024), “Impact of a deep learning sepsis prediction model on quality of care and survival,” NPJ Digital Medicine 7:14, doi:10.1038/s41746-023-00986-6. — Retrieved from: _____
2. Shashikumar, S. P., Wardi, G., Malhotra, A., & Nemati, S. (2021), “Artificial intelligence sepsis prediction algorithm learns to say ‘I don’t know,’” NPJ Digital Medicine 4:134 — the methodological-foundation paper for the conformal-prediction abstention structure. — Retrieved from: _____
3. Wong, A., Otles, E., Donnelly, J. P., Krumm, A., McCullough, J., DeTroyer-Cooley, O., et al. (2021), “External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients,” JAMA Internal Medicine 181(8):1065–1070 — the load-bearing external-validation paper on Epic Sepsis (Case 5). — Retrieved from: _____
4. Adams, R., Henry, K. E., Sridharan, A., Soleimani, H., Zhan, A., Rawat, N., Johnson, L., et al. (2022), “Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis,” Nature Medicine 28:1455–1460 — the prospective-positive TREWS deployment paper (Case 20). — Retrieved from: _____

CASE 35

Radiology AI Miscalibration

2018 – present

1. Larrazabal, A. J., Nieto, N., Peterson, V., Milone, D. H., & Ferrante, E. (2020), “Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis,” PNAS 117(23):12592–12594 — sensitivity drops for under-represented groups on NIH ChestX-ray14 and CheXpert. — Retrieved from: _____
2. Seyyed-Kalantari, L., Zhang, H., McDermott, M. B. A., Chen, I. Y., & Ghassemi, M. (2021), “Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations,” Nature Medicine 27:2176–2182 — disparities across Black, Hispanic, female, and lower-socioeconomic subgroups; persistence across model architectures. — Retrieved from: _____
3. Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019), “Dissecting racial bias in an algorithm used to manage the health of populations,” Science 366(6464):447–453 — proxy-label bias in a care-management algorithm; 200 million Americans/year are subject to tools of this class. — Retrieved from: _____
4. Wachter, R. M. & Brynjolfsson, E. (2023), “Will Generative Artificial Intelligence Deliver on Its Promise in Health Care?” JAMA 331(1):65–69 — generative-AI extension of the proxy-and-population problem. — Retrieved from: _____

5. FDA, “Proposed Regulatory Framework for Modifications to AI/ML-Based Software as a Medical Device” (2019); FDA draft guidance on AI/ML-Based SaMD (2025), with Predetermined Change Control Plan and Good Machine Learning Practice principles — the regulatory trajectory; absence of mandatory demographic stratification at clearance and post-market monitoring of in-use performance. — Retrieved from: _____

CASE 36 AlphaFold — Protein Structure Prediction

2020 – present

1. Jumper et al. (2021), “Highly accurate protein structure prediction with AlphaFold,” Nature — the method and accuracy. — Retrieved from: _____
2. Varadi et al. (2024), “AlphaFold Protein Structure Database in 2024,” Nucleic Acids Research — the 200M+ structures and open release. — Retrieved from: _____
3. Moult, J. (CASP organizer) commentary on AlphaFold2 (2020) — the benchmark and the achievement (paraphrased). — Retrieved from: _____
4. CASP benchmark documentation — the decades-long agreed evaluation. — Retrieved from: _____
5. Hassabis (DeepMind) public commentary — the open-release governance decision. — Retrieved from: _____

CASE 37 DeepMind Mammography — High-Profile Nature Paper, Replicability Push-back

2020

1. McKinney, S. M., Sieniek, M., Godbole, V., Godwin, J., Antropova, N., Ashrafian, H., Back, T., et al. (2020), “International evaluation of an AI system for breast cancer screening,” Nature 577:89–94, doi:10.1038/s41586-019-1799-6. — Retrieved from: _____
2. Haibe-Kains, B., Adam, G. A., Hosny, A., Khodakarami, F., MAQC Society Board of Directors, Waldron, L., Wang, B., et al. (2020), “Transparency and reproducibility in artificial intelligence,” Nature 586:E14–E16, doi:10.1038/s41586-020-2766-y. — Retrieved from: _____
3. McKinney et al. (2020), reply to Haibe-Kains et al., Nature 586:E17–E18 — the developers’ response on the reproducibility–infrastructure question. — Retrieved from: _____
4. Freeman, K., Geppert, J., Stinton, C., Todkill, D., Johnson, S., Clarke, A., & Taylor-Phillips, S. (2021), “Use of artificial intelligence for image analysis in breast cancer screening programmes: systematic review of test accuracy,” BMJ 374:n1872 — independent systematic review of subsequent AI-screening-deployment evidence. — Retrieved from: _____

CASE 38 iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms

2006 – 2011

1. Shin, J., Cheetham, T. C., Wong, L., Niu, F., Kass, E., Yoshinaga, M. A., Sorel, M., McCombs, J. S., & Sidney, S. (2011), “The impact of the iPLEDGE program on isotretinoin fetal exposure in an integrated health care system,” Journal of the American Academy of Dermatology, PMID:21565419. — Retrieved from: _____
2. FDA, iPLEDGE program documentation (2006 – present) — REMS architecture and enrollment requirements. — Retrieved from: _____
3. Collins, M. K., Moreau, J. F., Opel, D., Swan, J., Prevost, N., Hastings, M., Schwarz, E. B., & Ferris, L. K. (2014), “Compliance with pregnancy prevention measures during isotretinoin therapy,” Journal of the American Academy of Dermatology, 70(1):55–59, PMID:24157382 — source of the 150 annual exposures figure. — Retrieved from: _____
4. Sullivan et al. (2003), House Science Committee statement on SUBSAFE — the structural counterpoint (Case 173). — Retrieved from: _____

CASE 39 TeamSTEPPS

2006 – present

1. AHRQ, TeamSTEPPS 3.0 Curriculum (2023) — the framework and four competencies. — Retrieved from: _____
2. DoD / AHRQ partnership documentation — the joint development and implementation infrastructure. — Retrieved from: _____
3. Salas, E., Rosen, M. et al. (2009) — cross-domain team-training evidence base. — Retrieved from: _____
4. Weaver, S., Dy, S. & Rosen, M. (2014) — patient-safety team-training implementation and outcomes. — Retrieved from: _____
5. American Hospital Association Team Training Center — adoption and on-time-start metrics. — Retrieved from: _____

CASE 40 Team Science Training for Clinical and Translational Scientists

2020 – 2022

1. Colorado CTSA, “Team science training for clinical and translational scientists: An assessment of effectiveness,” Journal of Clinical and Translational Science, PMC12392353. — Retrieved from: _____
2. Falk-Krzesinski et al. (2010), “Mapping a research agenda for the science of team science,” Research Evaluation — broader team-science literature backdrop. — Retrieved from: _____
3. National Research Council (2015), Enhancing the Effectiveness of Team Science — the IOM/National Academies team-science synthesis. — Retrieved from: _____
4. v2 paired cases: IPE evidence gap (122), Implementation Science Training (123) — the frontier/measurement companions. — Retrieved from: _____

CASE 41 Implementation-Science Training — Stated Goals Outrunning Operational Practice

2020s

1. CTSA T32/TL1 program-goals study (2021), Journal of Clinical and Translational Science, PMC8826009. — Retrieved from: _____
2. Morris, Wooding, & Grant (2011), “The answer is 17 years, what is the question: understanding time lags in translational research,” Journal of the Royal Society of Medicine — the original 17-year-gap source for Case 13. — Retrieved from: _____
3. Brownson, Colditz, & Proctor (2018), Dissemination and Implementation Research in Health (2nd ed.) — the broader implementation-science synthesis. — Retrieved from: _____
4. v2 paired cases: Team-science training (121), IPE evidence gap (122). — Retrieved from: _____

CASE 42 Australian Hospital-Pharmacy Technician Role Redesign

2016

1. SHPA / Australian hospital-pharmacy network (2016), “Pharmacy Technician and Assistant Role Redesign within Australian Hospitals Project,” outcomes report. — Retrieved from: _____
2. Anderson et al. (2021), “Perceptions of hospital pharmacists and pharmacy technicians towards expanding roles for hospital pharmacy technicians: a cross-sectional survey,” Journal of Pharmacy Practice and Research, doi:10.1002/jppr.1697. — Retrieved from: _____
3. Boughen, Sutton, Fenn, & Wright (2017), “Defining the Role of the Pharmacy Technician and Identifying Their Future Role in Medicines Optimisation,” Pharmacy 5(3):40 — UK companion analysis. — Retrieved from: _____
4. Royal Pharmaceutical Society and SHPA practice statements on technician scope expansion — practitioner-tier framing the project sits inside. — Retrieved from: _____

CASE 43

Rwanda mHealth Maternal Care — Community Health Workers as the Capability Interface

2013 – 2018

1. Hategeka, C., Ruton, H., Law, M. R., et al. (2019), “Effect of a community health worker mHealth monitoring system on uptake of maternal and newborn health services in Rwanda,” *Global Health Research and Policy*, PMC6429813. — Retrieved from: _____
2. Rwanda Ministry of Health, community health program documentation and CHW scope-of-practice guidance, 2013–2018. — Retrieved from: _____
3. MIT News (2022), reporting on subsequent AI-augmented maternal-care work in Rwanda — journalism-tier companion to the peer-reviewed evaluation. — Retrieved from: _____
4. Cross-reference: Case 201 (PEPFAR HIV training-modality comparison) for the paired Sub-Saharan workforce-capability evidence. — Retrieved from: _____

CASE 44

Japan PMDA — Pre-Approved Change Management as Regulatory Architecture for AI/SaMD

2014 – present

1. Kikuchi, et al. (2025), “Scoping Review of Regulatory Transparency in AI-based Radiology Software: Analysis of PMDA-approved SaMD Products,” *medRxiv* 2025.10.02.25336333. — Retrieved from: _____
2. Aoki, T., et al. (2021 → published), “Regulatory-approved Deep Learning/Machine Learning-Based Medical Devices in Japan as of 2020: A Systematic Review,” *PMC9931274*. — Retrieved from: _____
3. Pharmaceuticals and Medical Devices Agency of Japan, PMD Act amendment (2019) and DASH for SaMD program documentation. — Retrieved from: _____
4. Cross-reference: “A decade of review in global regulation and research of artificial intelligence medical devices (2015–2025),” *PMCI2310608* — comparative regulatory context. — Retrieved from: _____

CASE 45

Tennessee Voluntary Pre-K Study

2009–2018

1. M. Lipsey, D. Farran & K. Durkin, “Effects of the Tennessee Pre-Kindergarten Program... Through Third Grade,” *Early Childhood Research Quarterly* 45: 155–176 (2018) — the RCT and fade-out. — Retrieved from: _____
2. K. Durkin, M. Lipsey, D. Farran & E. Wiesen, “Effects of a Statewide Pre-Kindergarten Program... Through Sixth Grade,” *Developmental Psychology* 58(3): 470–484 (2022) — the sixth-grade reversal (quoted). — Retrieved from: _____
3. Responses and counter-analyses to the TN-VPK findings (2018–2022) — the contested reception. — Retrieved from: _____
4. National Institute for Early Education Research, *State of Preschool yearbooks* (2010–2022) — program scale and context. — Retrieved from: _____
5. D. Phillips et al., *Puzzling It Out: The Current State of Scientific Knowledge on Pre-Kindergarten Effects* (2017); J. Heckman on durable early-childhood programs. — Retrieved from: _____

CASE 46

Algorithmic Bias in Educational Predictive Analytics

ongoing

1. Surveys of predictive-analytics adoption in U.S. higher education — a large majority of public colleges now use some form. — Retrieved from: _____
2. K. Bird et al., “Are Algorithms Biased in Education? Exploring Racial Bias in Predicting Community College Student Success,” *Journal of Policy Analysis and Management* 44 (2025), 379–402 — racial calibration bias, 5× higher at the bottom decile depending on the “at-risk” construct. — Retrieved from: _____

3. D. Gándara, H. Anahideh, M. Ison & L. Picchiarini, “Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student–Success Prediction,” AERA Open (2024) — bias traced to training data encoding historical discrimination. — Retrieved from: _____

4. R. Baker & A. Hawn, “Algorithmic Bias in Education,” International Journal of Artificial Intelligence in Education (2021) — foundational review of algorithmic bias in education. — Retrieved from: _____

5. Analyses of deficit framing and the interpretation of “at-risk” flags by faculty. — Retrieved from: _____

6. Cf. UK A-Level / Ofqual (Case 49); V. Eubanks, Automating Inequality (2018). — Retrieved from: _____

CASE 47 Algorithmic Bias in Automated Exam Proctoring 2022

1. Yoder–Himes, D. R., Asif, A., Kinney, K., Brandt, T. J., Cecil, R. E., Himes, P. R., Cashon, C., Hopp, R. M. P., & Ross, E. (2022). Racial, skin tone, and sex disparities in automated proctoring software. *Frontiers in Education*, 7:881449. doi:10.3389/feduc.2022.881449 — the case’s primary study. — Retrieved from: _____

2. Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. *Proceedings of Machine Learning Research*, 81:77–91 — the foundational intersectional-bias finding in face recognition. — Retrieved from: _____

3. Raji, I. D., & Buolamwini, J. (2019). Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial AI products. AAAI/ACM Conference on AI, Ethics, and Society — the audit-and-disclosure mechanism the case calls for. — Retrieved from: _____

4. Sjoding, M. W., Dickson, R. P., Iwashyna, T. J., Gay, S. E., & Valley, T. S. (2020). Racial bias in pulse oximetry measurement. *New England Journal of Medicine*, 383(25):2477–2478 — the structural analog in the medical-device context (Case 26). — Retrieved from: _____

CASE 48 School Surveillance and Black Student Outcomes — Infrastructure as the Mechanism 2010s – 2022

1. Johnson and colleagues (2022), school-surveillance infrastructure and Black student outcomes, *Journal of Criminal Justice* — the load-bearing peer-reviewed source for the case. — Retrieved from: _____

2. Hoffman, Trawalter, Axt, & Oliver (2016), “Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites,” *PNAS* 113(16):4296–4301 — race-construct trio at the cognitive-baseline layer (Case 6). — Retrieved from: _____

3. Inker, Eneanya, Coresh, et al. (2021), “New Creatinine- and Cystatin C–Based Equations to Estimate GFR without Race,” *NEJM* — race-construct trio at the formula layer (Case 25). — Retrieved from: _____

4. Sjoding, Dickson, Iwashyna, Gay, & Valley (2020), “Racial Bias in Pulse Oximetry Measurement,” *NEJM* 383:2477–2478 — race-construct trio at the device layer (Case 26). — Retrieved from: _____

CASE 49 UK Ofqual A-Level Algorithm — National-Scale Grading Replaced by Algorithm, Withdrawn in Days 2020

1. Ofqual, Awarding GCSE, AS, A level, advanced extension awards and extended project qualifications in summer 2020: interim report, August 2020. — Retrieved from: _____

2. UK House of Commons Education Committee, Getting the grades they’ve earned: COVID-19: the cancellation of exams and “calculated” grades, HC 254, July 2021. — Retrieved from: _____

3. Royal Statistical Society, Submission to the Office for Statistics Regulation on the summer 2020 grading process, 2020 — independent statistical review of the standardisation methodology. — Retrieved from: _____

4. Smith, H. (2020), “Algorithmic bias: should students pay the price?” *AI & Society* 35(4):1077–1078 — early academic commentary on the equity dimensions of the withdrawal. — Retrieved from: _____

CASE 50 Wisconsin DEWS — A Decade of Algorithmic Dropout Prediction

2012 – 2024

1. Perdomo, J. C., Britton, T., Hardt, M., & Abebe, R. (2025), "Difficult Lessons on Social Prediction from Wisconsin Public Schools," Proceedings of FAccT 2025, doi:10.1145/3715275.3732175 (also arXiv:2304.06205). — Retrieved from: _____
2. Wisconsin Department of Public Instruction (2021), Is DEWS Fair? — internal equity audit of the Dropout Early Warning System. — Retrieved from: _____
3. Feathers, T. (2023), "False Alarm: How Wisconsin Uses Race and Income to Label Students 'High Risk,'" The Markup, April 27, 2023 — investigation documenting the disparate-impact finding and the agency response. — Retrieved from: _____
4. Knowles, J. E. (2015), "Of Needles and Haystacks: Building an Accurate Statewide Dropout Early Warning System in Wisconsin," Journal of Educational Data Mining 7(3):18–67 — the original DEWS technical-methodology paper. — Retrieved from: _____

CASE 51 Atlanta Public Schools Cheating Scandal

2009 – 2015

1. Special Investigators, Final Report on Atlanta Public Schools (2011) — the organized cheating and the quoted finding. — Retrieved from: _____
2. Atlanta Journal-Constitution investigative series (2009–2011) — the erasure-rate analysis. — Retrieved from: _____
3. State of Georgia v. Hall et al. (2013–2015) — indictments and RICO convictions. — Retrieved from: _____
4. Koretz, D. (2017), The Testing Charade — high-stakes-testing distortion. — Retrieved from: _____
5. Campbell, D. (1976), "Assessing the Impact of Planned Social Change" — Campbell's Law. — Retrieved from: _____

CASE 52 Purdue Course Signals — The Reverse-Causality Retention Claim

2012 – 2013

1. Arnold, K. E., & Pistilli, M. D. (2012). Course Signals at Purdue: Using learning analytics to increase student success. Proceedings of LAK 2012, 267–270. doi:10.1145/2330601.2330666 — the original study at the center of the critique. — Retrieved from: _____
2. Caulfield, M. (2013). A discussion of the Purdue Course Signals retention numbers. e-Literate blog series and Inside Higher Ed commentary — the reverse-causality critique. — Retrieved from: _____
3. Essa, A. (2013). Simulation reproducing the Course Signals retention curve using a placebo treatment ("received a piece of chocolate") — methodological demonstration of the artifact. — Retrieved from: _____
4. Feldstein, M. (2013). What the Course Signals story says about learning-analytics evidence. e-Literate — the broader field-level critique. — Retrieved from: _____

CASE 53 inBloom

2014

1. M. Bulger, P. McCormick & M. Pitcan, The Legacy of inBloom, Data & Society Research Institute (2017) — inBloom as a failure of technocratic reform. — Retrieved from: _____
2. Education Week and Hechinger Report coverage of the state withdrawals (2013–2014) — nine states exiting. — Retrieved from: _____
3. Bulger et al. (2017) — governance, not technology, as the cause; "the technology was not the problem." — Retrieved from: _____
4. N. Selwyn, Distrusting Educational Technology (2014); d. boyd & K. Crawford, "Critical Questions for Big Data" (2012). — Retrieved from: _____

5. Parent Coalition for Student Privacy archives and the wave of state student-data-privacy legislation that followed inBloom. — Retrieved from: _____

CASE 54 Summit Learning / Personalized Learning Rollout

2014–2019

1. N. Bowles, “Silicon Valley Came to Kansas Schools. That Started a Rebellion,” New York Times (2019) — the parent revolt. — Retrieved from: _____
2. N. Singer, “The Silicon Valley Billionaires Remaking America’s Schools,” New York Times (2017) — the CZI/Summit rollout. — Retrieved from: _____
3. B. Herold, Education Week coverage of Summit / CZI implementation (2019) — district adoptions and withdrawals. — Retrieved from: _____
4. A. Watters, “The Stories We Tell About Personalized Learning,” Hack Education (2019) — the governance critique. — Retrieved from: _____
5. Chan Zuckerberg Initiative & Summit Learning public program documentation (2015–2019); cf. inBloom (Case 53). — Retrieved from: _____

CASE 55 Enrollment–Algorithm Yield Optimization Across U.S. Higher Education

2010s – present (Brookings synthesis 2021)

1. Engler, A. (2021), “Enrollment algorithms are contributing to the crises of higher education,” Brookings Institution, 14 Sept 2021. — Retrieved from: _____
2. Bettinger, E. et al. (2019), “Merit aid and college completion,” literature reviewed in Engler — graduation-odds effect sizes for \$1,000 of additional aid. — Retrieved from: _____
3. Goldrick-Rab, S. (2016), Paying the Price — broader synthesis on net-price, unsubsidized loans, and low-income completion. — Retrieved from: _____
4. Vendor case studies cited in Engler (Othot, EAB, Ruffalo Noel Levitz, University of Washington) — vendor-reported and not externally audited; flagged at evidence-tier under the title. — Retrieved from: _____

CASE 57 GAO Online Program Manager Oversight Gap (GAO–22–104463)

2022

1. U.S. Government Accountability Office (2022), “Higher Education: Education Needs to Strengthen Its Approach to Monitoring Colleges’ Arrangements with Online Program Managers,” GAO–22–104463. — Retrieved from: _____
2. U.S. Department of Education (2011), “Dear Colleague” guidance on incentive-compensation bundled-services exemption — the regulatory artifact the GAO audits. — Retrieved from: _____
3. Higher Education Act of 1992, incentive-compensation prohibition — the statutory basis the 2011 guidance interpreted. — Retrieved from: _____
4. 2U Inc. (2024), Chapter 11 bankruptcy filing (July 25, 2024); U.S. Department of Education, “Dear Colleague” Letter (Jan. 14, 2025) reaffirming the 2011 bundled-services guidance (GEN-11-05) — the commercial-boundary closure alongside the federal guidance’s continued validity. — Retrieved from: _____

CASE 58 USC × 2U Online MSW — When the Delegation Becomes the Product (Luna v. USC)

2010s – 2024

1. Stephanie Luna v. University of Southern California, class action complaint, Los Angeles County Superior Court, May 2023. — Retrieved from: _____
2. Higher Ed Dive reporting on the Luna complaint, May 2023; classaction.org and topclassactions.com summaries. — Retrieved from: _____

3. Project on Predatory Student Lending statement on USC–2U partnership termination, November 2023. — Retrieved from: _____
4. 2U Inc. and USC public statements on partnership termination, November 2023; broader 2U commercial-collapse reporting referenced through Case 57. — Retrieved from: _____

CASE 60 Houston EVAAS — Value-Added Teacher Evaluation on Trial

2011–2017

1. Hous. Fed’n of Teachers v. Hous. Indep. Sch. Dist., 251 F. Supp. 3d 1168 (S.D. Tex. 2017) — memorandum opinion denying summary judgment on the procedural due process claim. — Retrieved from: _____
2. Amrein–Beardsley, A., & Collins, C. (2012), “The SAS Education Value-Added Assessment System (SAS® EVAAS®) in the Houston Independent School District (HISD): Intended and Unintended Consequences,” Education Policy Analysis Archives 20(12), doi:10.14507/epaa.v20n12.2012. — Retrieved from: _____
3. American Statistical Association (2014), ASA Statement on Using Value-Added Models for Educational Assessment, April 8, 2014. — Retrieved from: _____
4. American Educational Research Association Council (2015), “AERA Statement on Use of Value-Added Models (VAM) for the Evaluation of Educators and Educator Preparation Programs,” Educational Researcher 44(8):448–452, doi:10.3102/0013189X15618385. — Retrieved from: _____
5. Amrein–Beardsley, A., & Geiger, T. (2020), “Methodological Concerns About the Education Value-Added Assessment System (EVAAS): Validity, Reliability, and Bias,” SAGE Open 10(2), doi:10.1177/2158244020922224. — Retrieved from: _____
6. Education Week (October 2017), “Houston District Settles Lawsuit With Teachers’ Union Over Value-Added Scores”; Houston Public Media (October 10, 2017), “Federal Lawsuit Settled Between Houston’s Teacher Union and HISD” — settlement terms and the \$237,000 fee figure. — Retrieved from: _____

CASE 61 Sold a Story — The Science of Reading and the Decades-Long Research-Practice Gap

1997–2023

1. Hanford, E. (2022 – 2024), “Sold a Story: How Teaching Kids to Read Went So Wrong,” APM Reports, podcast series launched October 20, 2022. — Retrieved from: _____
2. Hanford, E. (2018), “Hard Words: Why Aren’t Our Kids Being Taught to Read?” APM Reports, September 10, 2018. — Retrieved from: _____
3. National Reading Panel (2000), Teaching Children to Read: An Evidence-Based Assessment of the Scientific Research Literature on Reading and Its Implications for Reading Instruction, National Institute of Child Health and Human Development, NIH Pub. No. 00–4769. — Retrieved from: _____
4. Castles, A., Rastle, K., & Nation, K. (2018), “Ending the Reading Wars: Reading Acquisition From Novice to Expert,” Psychological Science in the Public Interest 19(1):5 – 51. — Retrieved from: _____
5. Schwartz, S. (updated through 2024), “Which States Have Passed ‘Science of Reading’ Laws? What’s in Them?” Education Week — legislation tracker; 40 states plus DC by late 2024. — Retrieved from: _____
6. Seidenberg, M. (2017), Language at the Speed of Sight: How We Read, Why So Many Can’t, and What Can Be Done About It, Basic Books. — Retrieved from: _____

CASE 62 Gates Intensive Partnerships — Measurement Without a Coupled Lever

2009–2018

1. Stecher, B. M., Holtzman, D. J., Garet, M. S., Hamilton, L. S., et al. (2018), Improving Teaching Effectiveness: Final Report: The Intensive Partnerships for Effective Teaching Through 2015–2016, RAND Corporation, RR–2242–BMGF. — Retrieved from: _____

2. Kane, T. J., McCaffrey, D. F., Miller, T., & Staiger, D. O. (2013), Have We Identified Effective Teachers? Validating Measures of Effective Teaching Using Random Assignment, MET Project research paper, Bill & Melinda Gates Foundation, January 2013. — Retrieved from: _____
3. Sokol, M. (2015), “Sticker shock: How Hillsborough County’s Gates grant became a budget buster,” Tampa Bay Times, October 2015 — with companion reporting, “Hillsborough schools to dismantle Gates-funded system that cost millions to develop.” — Retrieved from: _____
4. Stecher, B. M., et al. (2016), Improving Teaching Effectiveness: Implementation: The Intensive Partnerships for Effective Teaching Through 2013–2014, RAND Corporation, RR-1295. — Retrieved from: _____
5. Sawchuk, S. (2018), “‘An Expensive Experiment’: Gates Teacher-Effectiveness Program Shows No Gains for Students,” Education Week, June 21, 2018. — Retrieved from: _____

CASE 63 LAUSD iPad Program — The Common Core Technology Project

2013–2015

1. Gilbertson, A. (2014), “LA schools cancel iPad contracts after KPCC publishes internal emails,” KPCC 89.3 (Southern California Public Radio), August 25, 2014 — the public-records email investigation and the contract halt; see also KPCC’s “LAUSD iPads: Timeline of a troubled program.” — Retrieved from: _____
2. American Institutes for Research (2014), interim evaluation of the LAUSD Common Core Technology Project, Phase 1 — 245 classroom observations across CCTP schools; Pearson curriculum observed in use in one. — Retrieved from: _____
3. Herold, B. (2014), “Hard Lessons Learned in Ambitious L.A. iPad Initiative,” Education Week, September 2014 — retrospective on the implementation failures; see also Herold (2014), “FBI Investigation Leaves L.A. iPad Initiative in Further Disarray,” Education Week, December 2014. — Retrieved from: _____
4. Blume, H., Los Angeles Times coverage of the CCTP (2013–2015) — the rollout, the security bypass, the contract suspension, the Deasy resignation, and the FBI document seizure. — Retrieved from: _____
5. Education Week Market Brief (2017), “Feds Drop Investigation Into Los Angeles District Over \$1 Billion iPad Purchase,” February 2017 — the reported closure of the federal inquiry without charges. — Retrieved from: _____

CASE 65 Training Transfer — The Gap Between Learning and Doing

2010

1. Blume, Ford, Baldwin, & Huang (2010), “Transfer of Training: A Meta-Analytic Review,” Journal of Management 36(4):1065–1105, doi:10.1177/0149206309352880. — Retrieved from: _____
2. Baldwin & Ford (1988), “Transfer of Training: A Review and Directions for Future Research,” Personnel Psychology 41(1):63–105 — the foundational synthesis the 2010 meta-analysis updates. — Retrieved from: _____
3. Burke & Hutchins (2007), “Training Transfer: An Integrative Literature Review,” Human Resource Development Review 6(3):263–296 — the integrative-review companion synthesis. — Retrieved from: _____
4. Kirkpatrick & Kirkpatrick (2006), Evaluating Training Programs — the framework the meta-analysis informs (paired Case 79). — Retrieved from: _____

CASE 66 Growth-Mindset National Experiment — Heterogeneous Effects

2019

1. Yeager, D. S., Hanselman, P., Walton, G. M., Murray, J. S., Crosnoe, R., Muller, C., Tipton, E., Schneider, B., Hulleman, C. S., Hinojosa, C. P., Paunesku, D., Romero, C., Flint, K., Roberts, A., Trott, J., Iachan, R., Buontempo, J., Yang, S. M., Carvalho, C. M., Hahn, P. R., Gopalan, M., Mhatre, P., Ferguson, R., Duckworth, A. L., & Dweck, C. S. (2019). A national experiment reveals where a growth mindset improves achievement. Nature, 573(7774):364–369. doi:10.1038/s41586-019-1466-y — the case’s primary trial. — Retrieved from: _____
2. National Study of Learning Mindsets, ICPSR 37353 — the trial dataset. — Retrieved from: _____
3. Dweck, C. S. (2006). Mindset: The New Psychology of Success. Random House — the broader theoretical framework the intervention rests on. — Retrieved from: _____

4. Sisk, V. F., Burgoyne, A. P., Sun, J., Butler, J. L., & Macnamara, B. N. (2018). To what extent and under which circumstances are growth mindsets important to academic achievement? Two meta-analyses. *Psychological Science*, 29(4):549–571 — the prior moderator-analysis literature the Yeager trial extends. — Retrieved from: _____

CASE 67 Cognitive Tutor / Carnegie Learning

1990s – present

1. Anderson, J., Corbett, A., Koedinger, K. & Pelletier, R. (1995), “Cognitive Tutors: Lessons Learned,” *Journal of the Learning Sciences* — the ACT-R basis and design. — Retrieved from: _____
2. Koedinger, K. & Corbett, A. (2006), *Cambridge Handbook of the Learning Sciences* — knowledge tracing and adaptive instruction. — Retrieved from: _____
3. RAND Corporation Algebra I evaluation — the statistically significant achievement effects. — Retrieved from: _____
4. Carnegie Learning program documentation — scale to 3,000-plus schools. — Retrieved from: _____
5. Aleven, V. et al. (2016), example-tracing tutors — the limits in ill-structured domains. — Retrieved from: _____

CASE 68 CIRCUIT — A Scalable, Equity-Centered Research Workforce Model

2017 – 2023 (six cycles)

1. Cervantes, Floryanzia, Sharp, Gray-Roncal, & Johnson (2023), “Empowering Trailblazers toward Scalable, Systematized, Research-Based Workforce Development,” *ASEE Annual Conference*, doi:10.18260/1-2-43271. — Retrieved from: _____
2. CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional program description. — Retrieved from: _____
3. National Academies / NSF research on undergraduate research experience effects on STEM persistence (broader literature against which the case sits). — Retrieved from: _____
4. Paired case (78) — CIRCUIT proofreading + MICrONS — the deployed-capability companion. — Retrieved from: _____

CASE 69 Duolingo Half-Life Regression — Spaced Repetition at Consumer Scale

2016

1. Settles, B., & Meeder, B. (2016), “A Trainable Spaced Repetition Model for Language Learning,” *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, pp. 1848–1858, doi:10.18653/v1/P16-1174. — Retrieved from: _____
2. Ebbinghaus, H. (1885), *Über das Gedächtnis* — the empirical forgetting-curve foundation HLR formalizes. — Retrieved from: _____
3. Pimsleur, P. (1967), “A memory schedule,” *Modern Language Journal* 51(2):73–75 — graduated-interval recall heuristic. — Retrieved from: _____
4. Leitner, S. (1972), *So lernt man lernen* — Leitner-box spacing heuristic. — Retrieved from: _____

CASE 70 High-Impact Learning System — Engineering the Environment, Not Just the Event

2001 – present

1. Brinkerhoff, R. O., & Apking, A. M. (2001), *High Impact Learning: Strategies for Leveraging Performance and Business Results from Training Investments*, Basic Books. — Retrieved from: _____
2. Watershed LRS practitioner summaries of HILS deployment patterns. — Retrieved from: _____
3. L-TEN (Life Sciences Trainers and Educators Network) practitioner summaries of HILS in life-sciences L&D. — Retrieved from: _____

4. Blume et al. (2010), *Journal of Management* 36(4):1065–1105 — the meta-analytic finding HILS operationalizes (paired Case 65). — Retrieved from: _____
5. Brinkerhoff (2005), *Advances in Developing Human Resources* 7(1):86–101 — SCM as the paired evaluation instrument (Case 83). — Retrieved from: _____

CASE 71 **Reflective Practice on a Work-Based Software Engineering Program** 2025 (preprint) — Longitudinal Capability Development

1. “The Development of Reflective Practice on a Work-Based Software Engineering Program: A Longitudinal Study,” arXiv:2504.20956 (2025) — preprint. — Retrieved from: _____
2. D. Schön, *The Reflective Practitioner* (1983) — the foundational account of reflection-in-action the genre rests on. — Retrieved from: _____
3. Boud, Keogh & Walker (eds.), *Reflection: Turning Experience into Learning* (1985) — reflection as a learning process, and the measurement problem it raises. — Retrieved from: _____
4. the proposed revisions — the amended LEO-2 (first-person narration of design iteration) the case evaluates. — Retrieved from: _____

CASE 72 **ASSISTments — National Replication and Long-Term Follow-Through** 2014 – present

1. Roschelle, J., Feng, M., Murphy, R. F., & Mason, C. A. (2016), “Online Mathematics Homework Increases Student Achievement,” *AERA Open* 2(4):1–12, doi:10.1177/2332858416673968. — Retrieved from: _____
2. Murphy, R. et al. (2020), follow-up evaluation extending the 7th-grade effect into 8th-grade outcomes. — Retrieved from: _____
3. Heffernan, N. T., & Heffernan, C. L. (2014), “The ASSISTments ecosystem,” *International Journal of AI in Education* 24:470–497 — platform design and research-loop description. — Retrieved from: _____
4. Arnold Ventures RCT documentation of the virtual-training-adaptation arm — longitudinal follow-through through end of 8th grade. — Retrieved from: _____

CASE 73 **The Doer Effect at Scale — Replication, AI-Generated Questions, Non-WEIRD Extension** 2016 – 2025

1. Van Campenhout, R., Jerome, B., Dittel, J. S., & Johnson, B. G. (2023), “The Doer Effect at Scale: Investigating Correlation and Causation Across Seven Courses,” *LAK23*, doi:10.1145/3576050.3576103. — Retrieved from: _____
2. Van Campenhout et al. (2025), “Scaling the Doer Effect: A Replication Analysis Using AI-Generated Questions,” *L@S '25*, doi:10.1145/3698205.3729545. — Retrieved from: _____
3. Butler, D. et al. (2025), “Does the Doer Effect Generalize To Non-WEIRD Populations? Toward Analytics in Radio and Phone-Based Learning,” *LAK '25*, doi:10.1145/3706468.3706505 (also arXiv 2412.20923). — Retrieved from: _____
4. Koedinger, K. R. et al. (2016), original doer-effect causal-claim paper at LAK 2016 — the replication target the present case builds on. — Retrieved from: _____

CASE 74 **Zhang/Scardamalia — Knowledge Building Across Three Cohorts** 2009

1. Zhang, J., Scardamalia, M., Reeve, R., & Messina, R. (2009), “Designs for Collective Cognitive Responsibility in Knowledge-Building Communities,” *Journal of the Learning Sciences* 18(1):7–44, doi:10.1080/10508400802581676. — Retrieved from: _____
2. Scardamalia, M., & Bereiter, C. (2006), “Knowledge Building: Theory, pedagogy, and technology,” in K. Sawyer (ed.), *Cambridge Handbook of the Learning Sciences* — programmatic backdrop. — Retrieved from: _____

3. Bereiter, C., & Scardamalia, M. (2014), "Knowledge Building and Knowledge Creation: One Concept, Two Hills to Climb," in S. C. Tan, H. J. So, & J. Yeo (eds.), *Knowledge Creation in Education — extension into the post-2009 program*. — Retrieved from: _____
4. Design-based research methodology references — Cobb, Confrey, diSessa, Lehrer, & Schauble (2003), "Design Experiments in Educational Research," *Educational Researcher* 32(1):9–13 — the methodological frame the 2009 study operates inside. — Retrieved from: _____

CASE 75 **Chen et al. — Rural China AI Devices and the Equity-Direction Finding**

2025

1. Chen, R., Wu, Y., Chen, Z., & Zhou, P. (2025), "Advancing educational equity in rural China: the impact of AI devices on teaching quality and learning outcomes for sustainable development," *Frontiers in Psychology* 16:1588047, doi:10.3389/fpsyg.2025.1588047. — Retrieved from: _____
2. Paired Case 86 (Gándara et al., AERA Open) — community-college predictive equity at the higher-education scale. — Retrieved from: _____
3. Paired Case 88 (LiveHint AI bias across foundation models, AIED 2025) — bias-surfacing in AI-supported instruction. — Retrieved from: _____
4. Paired Case 73 (Doer Effect non-WEIRD LAK 2025 radio-and-phone extension) — non-WEIRD methodological discipline at the heterogeneity-finding axis. — Retrieved from: _____

CASE 76 **Multimodal Learning Analytics In-the-Wild — A First-Person Lessons-Learned Account**

2023

1. Martinez-Maldonado, R. et al. (2023), "Lessons Learnt from a Multimodal Learning Analytics Deployment In-the-Wild," arXiv:2303.09099 — preprint, sections published in adjacent LAK and IEEE TLT outlets. — Retrieved from: _____
2. Worsley, M., & Blikstein, P. (2018), "A multimodal multisensor framework for examining how students engage in design," *Journal of Learning Analytics* — broader MMLA literature backdrop. — Retrieved from: _____
3. Schon, D. (1983), *The Reflective Practitioner* — the genre's theoretical underpinning, referenced across the reflective-practice case tier. — Retrieved from: _____
4. Editors' memo (B1) — Practice Flywheel commissioning structure that the case is offered as a published-first-person exemplar within. — Retrieved from: _____

CASE 77 **Hybrid Human-AI Tutoring — Augmentation, Not Delegation**

2024

1. Thomas, D. R. et al. (2024), "Improving Student Learning with Hybrid Human-AI Tutoring: A Three-Study Quasi-Experimental Investigation," *LAK '24*, doi:10.1145/3636555.3636896. — Retrieved from: _____
2. Case 20 (TREWS) reference set — Henry et al. (2022), *Nature Medicine* — clinician-AI teaming analog. — Retrieved from: _____
3. Case 5 (Epic Sepsis) reference set — Wong et al. (2021), *JAMA Internal Medicine* — delegation-collapse analog. — Retrieved from: _____
4. Koedinger, K. R. et al. — Cognitive Tutor literature as the fully-automated track the augmentation track contrasts with. — Retrieved from: _____

CASE 78 **CIRCUIT / MICrONS — The Human Correction Layer at Petabyte Scale**

2017 – present

1. MICrONS program literature (IARPA) — connectomics method and automated segmentation evidence base. — Retrieved from: _____

2. APL BossDB documentation — petabyte-scale connectomics data infrastructure. — Retrieved from: _____
3. CIRCUIT program documentation (JHU Hub, 2017 – present) — institutional/program description; the training-outcome evidence is at this tier and the evidence-tier flag is binding. — Retrieved from: _____
4. Cervantes et al. (2023), ASEE Annual Conference — the paired peer-reviewed CIRCUIT case (Case 68). — Retrieved from: _____
5. Wachter & Brynjolfsson (2023), JAMA — generative AI verification framing, applicable across domains. — Retrieved from: _____

CASE 79 The Kirkpatrick Chain of Evidence — Where Corporate L&D Stops 1959 – present

1. Kirkpatrick (1959–1960), original four-level evaluation series in Journal of the ASTD; updated as Kirkpatrick & Kirkpatrick, Evaluating Training Programs (3rd ed., 2006). — Retrieved from: _____
2. Devlin Peck, “Kirkpatrick Model: A Guide to the Four Levels of Training Evaluation” — synthesis of the stop-at-L2 pattern in corporate practice. — Retrieved from: _____
3. D2L, “Kirkpatrick’s 4 Levels of Training Evaluation,” practitioner guide documenting the same pattern. — Retrieved from: _____
4. Valamis, “Kirkpatrick Model” practitioner guide on evaluation-practice gaps. — Retrieved from: _____
5. Blume, Ford, Baldwin, & Huang (2010), “Transfer of Training: A Meta-Analytic Review,” Journal of Management 36(4):1065–1105 — the paired peer-reviewed case (65). — Retrieved from: _____

CASE 80 Georgia State University Predictive Analytics 2012 – present

1. Renick, T. & Strom, A. (2020) on GSU’s advising transformation — the system design and outcomes. — Retrieved from: _____
2. Georgia State University institutional research and Strategic Plan reports — graduation-rate and equity data. — Retrieved from: _____
3. New York Times, “How Colleges Know You’re Not Finishing” (2018) — the 800-factor advising model. — Retrieved from: _____
4. EDUCAUSE Review on GSU predictive advising — the human-loop architecture. — Retrieved from: _____
5. Complete College America, Game Changers documentation — dissemination of the model. — Retrieved from: _____

CASE 81 Open University ‘Ethical Use of Student Data’ and OU Analyse 2014 – 2025

1. Slade & Prinsloo (2013), “Learning Analytics: Ethical Issues and Dilemmas,” American Behavioral Scientist 57(10):1510–1529, doi:10.1177/0002764213479366. — Retrieved from: _____
2. Open University (2014), “Ethical Use of Student Data for Learning Analytics” — first institutional policy of its kind in higher education. — Retrieved from: _____
3. Rienties, Boroowa, Cross, Kubiak, Mayles, & Murphy (2016), “Analytics4Action Evaluation Framework: A Review of Evidence-Based Learning Analytics Interventions at the Open University UK,” Journal of Interactive Media in Education 2016(1):2, doi:10.5334/jime.394. — Retrieved from: _____
4. Herodotou, Hlosta, Boroowa, Rienties, Zdrahal, & Mangafa (2019), “Empowering online teachers through predictive learning analytics,” British Journal of Educational Technology 50(6):3064–3079, doi:10.1111/bjet.12853 — OU Analyse evaluation across 559 teachers (189 with OUA access) and 14,000+ students in 15 undergraduate courses; average-use teachers benefited students most. — Retrieved from: _____

CASE 82 **MMALA — A Maturity Model for Responsible Learning–Analytics Adoption (Brazil)** 2024

1. Freitas, Fonseca, Garcia, Pontual Falcão, Marques, Gasevic, & Mello (2024), “MMALA: Developing and Evaluating a Maturity Model for Adopting Learning Analytics,” *Journal of Learning Analytics* 11(1):67–86. — Retrieved from: _____
2. Open University Ethical Use of Student Data policy (2014) — institutional-policy companion (Case 81). — Retrieved from: _____
3. Hilliger et al. (2020), *Internet and Higher Education* — LALA participatory adoption companion (Case 91). — Retrieved from: _____
4. Norwegian Expert Commission on Learning Analytics, final NOU (2023) — national-scale companion (Case 92). — Retrieved from: _____

CASE 83 **Brinkerhoff Success Case Method — Tails as the Evaluation Instrument** 2005 – present

1. Brinkerhoff, R. O. (2005), “The Success Case Method: A Strategic Evaluation Approach to Increasing the Value and Effect of Training,” *Advances in Developing Human Resources* 7(1):86–101, doi:10.1177/1523422304272172. — Retrieved from: _____
2. Brinkerhoff Evaluation Institute deployment list — Cargill, Ford, Merck, World Bank, International Committee of the Red Cross — practitioner channel. — Retrieved from: _____
3. Kirkpatrick & Kirkpatrick (2006), *Evaluating Training Programs* — the chain-of-evidence framework SCM operationalizes (paired Case 79). — Retrieved from: _____
4. Blume, Ford, Baldwin, & Huang (2010), *Journal of Management* 36(4):1065–1105 — the meta-analytic environment-as-decisive finding SCM samples around (paired Case 65). — Retrieved from: _____

CASE 84 **Cognitive Tutor Algebra I at Scale — Year-One Null, Year-Two Positive** 2007 – 2014

1. Pane, J. F., Griffin, B. A., McCaffrey, D. F., & Karam, R. (2014), “Effectiveness of Cognitive Tutor Algebra I at Scale,” *Educational Evaluation and Policy Analysis* 36(2):127–144, doi:10.3102/0162373713507480. — Retrieved from: _____
2. RAND Working Paper WR-1050 — addendum to the Pane et al. evaluation. — Retrieved from: _____
3. Koedinger, K. R., Anderson, J. R., Hadley, W. H., & Mark, M. A. (1997), “Intelligent tutoring goes to school in the big city,” *IJAIED* — the v1 Case 67 system description Cognitive Tutor builds from. — Retrieved from: _____
4. What Works Clearinghouse — Cognitive Tutor evidence-base summary applying federal-grade evidence standards to the Pane et al. trial. — Retrieved from: _____

CASE 85 **OU Analyse — Predictive Learning Analytics and Teacher Use at Scale** 2019 – 2023

1. Herodotou, C., Hlosta, M., Boroowa, A., Rienties, B., Zdrahal, Z., & Mangafa, C. (2019), “Empowering online teachers through predictive learning analytics,” *British Journal of Educational Technology* 50(6):3064–3079, doi:10.1111/bjet.12853. — Retrieved from: _____
2. Herodotou, C. et al. (2023), “Predictive Learning Analytics and University Teachers: Usage and perceptions three years post implementation,” *LAK '23*, doi:10.1145/3576050.3576061. — Retrieved from: _____
3. Herodotou et al. (2019), “A large-scale implementation of predictive learning analytics in higher education: the teachers’ role and perspective,” *Educational Technology Research and Development*, ERIC EJ1227972 — complementary teacher-perspective paper. — Retrieved from: _____

4. Arnold, K. E., & Pistilli, M. D. (2012), "Course Signals at Purdue," LAK '12 — the structural precursor v1 carries as a cautionary case. — Retrieved from: _____

CASE 86 **Gándara — Detecting and Mitigating Algorithmic Bias in College Student-Success Prediction** 2024

1. Gándara, Anahideh, Ison, & Picchiarini (2024), "Inside the Black Box: Detecting and Mitigating Algorithmic Bias across Racialized Groups in College Student-Success Prediction," AERA Open 10, doi:10.1177/23328584241258741 — primary case source on cross-group accuracy disparity, stratified evaluation, and bias mitigation in student-success prediction. — Retrieved from: _____
2. Barocas, Hardt, & Narayanan, Fairness and Machine Learning (fairmlbook.org) — methodological backdrop on construct definition and stratified evaluation. — Retrieved from: _____
3. Friedler, Scheidegger, & Venkatasubramanian (2021), "The (im)possibility of fairness: different value systems require different mechanisms for fair decision making," Communications of the ACM — the construct-definition argument at field level. — Retrieved from: _____
4. v2 cross-referenced cases: 25 (eGFR race correction), 26 (pulse oximetry across skin tones), 6 (Hoffman pain bias) — the race-construct trio at the construct-definition layer. — Retrieved from: _____

CASE 87 **Yu / Lee / Kizilcec — Protected Attributes in Learning-Analytics Models** 2021 – 2024

1. Yu, R., Lee, H., & Kizilcec, R. F. (2021), "Should College Dropout Prediction Models Include Protected Attributes?" in Proceedings of the Eighth ACM Conference on Learning @ Scale (L@S '21), doi:10.1145/3430895.3460139 — primary paper on the include-or-exclude empirical analysis. — Retrieved from: _____
2. Kizilcec & Lee, "Algorithmic Fairness in Education," in Holmes & Porayska-Pomsta (eds.), Ethics in Artificial Intelligence in Education — broader synthesis of the fairness-in-learning-analytics frontier. — Retrieved from: _____
3. Dwork, Hardt, Pitassi, Reingold, & Zemel (2012), "Fairness through awareness," Proceedings of ITCS — foundational paper on the inadequacy of fairness-through-unawareness. — Retrieved from: _____
4. Barocas, Hardt, & Narayanan, Fairness and Machine Learning (fairmlbook.org) — technical-fairness backdrop. — Retrieved from: _____
5. v2 cross-referenced cases: 86 (Gándara), 25 (eGFR), 26 (pulse oximetry), 6 (Hoffman) — equity-construct five-case set. — Retrieved from: _____

CASE 88 **LiveHint AI — Evaluating an AI Tutor for Bias Across Foundation Models** 2025

1. AIED 2025, "Evaluating an AI Tutor for Bias Across Different Foundation Models," Springer/ACM proceedings, doi:10.1007/978-3-031-98465-5_43; preprint at renzheyu.com/papers/AIED2025_Tutor.pdf. — Retrieved from: _____
2. Bommasani, R. et al. (2021), "On the Opportunities and Risks of Foundation Models," Stanford CRFM — the foundation-model framing the case builds on. — Retrieved from: _____
3. Race-construct trio reference set: Hoffman et al. (2016), Sjoding et al. (2020) pulse oximetry, Inker et al. (2021) eGFR-without-race — paired with Cases 25, 26, 6. — Retrieved from: _____
4. Carnegie Learning LiveHint product documentation — the subject system; case framing is binding on LiveHint being in development, not deployment. — Retrieved from: _____

CASE 89 Data Privacy and Learning Analytics on the African Continent

2022

1. Prinsloo, P., & Kaliisa, R. (2022), "Data privacy on the African continent: Opportunities, challenges and implications for learning analytics," *British Journal of Educational Technology* 53(4):894–913, doi:10.1111/bjet.13226. — Retrieved from: _____
2. African Union Convention on Cyber Security and Personal Data Protection (Malabo Convention, 2014) — continental policy backdrop, partial ratification. — Retrieved from: _____
3. Slade & Prinsloo (2013), *American Behavioral Scientist* — earlier framing on which the 2022 paper builds. — Retrieved from: _____
4. Open University Ethical Use of Student Data policy (2014) — single-regime governance-by-design companion. — Retrieved from: _____

CASE 90 Algorithmic College Admissions — Vendors' Claims vs. Applicants' Perceptions

2025

1. Pyle, C., & Andalibi, N. (2025), "Algorithmic College Admissions in the U.S.: Distances Between Vendors' Claims and Applicants' Perceptions," *Proceedings of the ACM on Human-Computer Interaction* 9(7), CSCW369, doi:10.1145/3757550. — Retrieved from: _____
2. Engler (2021), Brookings — paired deployment-side mapping (Case 55). — Retrieved from: _____
3. GAO-22-104463 (2022) — paired regulator-side audit (Case 57). — Retrieved from: _____
4. Dunne, A., & Raby, F. (2013), *Speculative Everything* — methodological backdrop for speculative-design probes. — Retrieved from: _____

CASE 91 LALA — Building Learning-Analytics Governance Capacity Across Latin America

2017 – 2020

1. Hilliger, Ortiz-Rojas, Pesántez-Cabrera, Scheihing, Tsai, Muñoz-Merino, Broos, Whitelock-Wainwright, & Pérez-Sanagustín (2020), "Identifying needs for learning analytics adoption in Latin American universities: A mixed-methods approach," *The Internet and Higher Education* 45:100726, doi:10.1016/j.iheduc.2020.100726. — Retrieved from: _____
2. LALA project — EU Erasmus+ grant 586120-EPP-1-2017-1-ES (2017–2020) — program documentation and the LALA CANVAS artifact. — Retrieved from: _____
3. Open University Ethical Use of Student Data policy (2014) and OU Analyse — UK companion governance-by-design case (Case 81). — Retrieved from: _____
4. Slade & Prinsloo (2013), "Learning Analytics: Ethical Issues and Dilemmas," *American Behavioral Scientist* 57(10):1510–1529 — the broader field-scale ethics framing. — Retrieved from: _____

CASE 92 Norway's National Expert Commission on Learning Analytics

2021 – 2023

1. Norwegian Expert Commission on Learning Analytics, interim report to the Minister of Education and Research (June 2022). — Retrieved from: _____
2. Norwegian Expert Commission on Learning Analytics, final NOU report (2023), *Norges offentlige utredninger*. — Retrieved from: _____
3. Misiejuk & Wasson (2023), "Learning analytics in Norway: A national perspective," *Journal of Learning Analytics* — secondary academic synthesis of the commission and its dilemmas. — Retrieved from: _____
4. Hilliger et al. (2020), *Internet and Higher Education* — the LALA companion at multi-country participatory scale (Case 91). — Retrieved from: _____

CASE 93 Singapore SkillsFuture — National Workforce Capability at Scale 2015 – present

1. SkillsFuture Singapore (SSG), Year-in-Review 2024 — program metrics and outcome reporting. — Retrieved from: _____

2. Ministry of Education (MOE), Singapore, parliamentary replies on TRAQOM, 2020. — Retrieved from: _____

3. International Labour Organization (ILO), “Investigating an Upskilling Programme in Singapore” — international comparative analysis. — Retrieved from: _____

4. “Future-Skilling the Workforce: SkillsFuture Movement in Singapore,” Springer, 2024 — peer-reviewed program analysis. — Retrieved from: _____

5. Ministry of Trade and Industry (MTI), Singapore, 2018 — WSQ wage-premium study. — Retrieved from: _____

CASE 94 Learning Analytics on the African Continent — A Scoping Review as the Present-State Map 2022

1. Prinsloo, P., & Kaliisa, R. (2022), “Learning Analytics on the African Continent: An Emerging Research Focus and Practice,” Journal of Learning Analytics; ResearchGate publication 361096718. — Retrieved from: _____

2. Lemmens, J.-C., & Henn, M. (2015), South African Association for Institutional Research (SAAIR) proceedings — adjacent SA higher-education learning-analytics work. — Retrieved from: _____

3. SciELO (2020), “Development of a contextualised learning-analytics framework for South African higher education.” — Retrieved from: _____

4. Cross-reference: the African data-privacy governance case earlier in the corpus, for the construct-travel problem stated in governance terms. — Retrieved from: _____

CASE 95 PBIS Implementation Fidelity — Why the Framework Travels Only with Its Measurement Loop 1997–2024

1. Bradshaw, Mitchell, & Leaf (2010), “Examining the Effects of Schoolwide Positive Behavioral Interventions and Supports on Student Outcomes: Results From a Randomized Controlled Effectiveness Trial in Elementary Schools,” Journal of Positive Behavior Interventions, 12(3), 133 – 148. — Retrieved from: _____

2. Bradshaw, Waasdorp, & Leaf (2012), “Effects of School-Wide Positive Behavioral Interventions and Supports on Child Behavior Problems,” Pediatrics, 130(5), e1136 – e1145. — Retrieved from: _____

3. Horner, Todd, Lewis-Palmer, Irvin, Sugai, & Boland (2004), “The School-Wide Evaluation Tool (SET): A Research Instrument for Assessing School-Wide Positive Behavior Support,” Journal of Positive Behavior Interventions, 6(1), 3 – 12. — Retrieved from: _____

4. McIntosh, Massar, Algozzine, George, Horner, Lewis, & Swain-Bradway (2017), “Technical Adequacy of the SWPBIS Tiered Fidelity Inventory,” Journal of Positive Behavior Interventions, 19(1), 3 – 13. — Retrieved from: _____

5. McIntosh, Mercer, Nese, Strickland-Cohen, & Hoselton (2016), “Predictors of Sustained Implementation of School-Wide Positive Behavioral Interventions and Supports,” Journal of Positive Behavior Interventions, 18(4), 209 – 218. — Retrieved from: _____

6. Bradshaw, Pas, et al. (2012), “A State-Wide Partnership to Promote Safe and Supportive Schools: The PBIS Maryland Initiative,” Administration and Policy in Mental Health, 39(4), 225 – 237. — Retrieved from: _____

CASE 96 AeroPerú Flight 603

1996

1. Peru Dirección General de Aeronáutica Civil, accident investigation board, final report on AeroPerú 603 (October 1997; with NTSB/FAA/Boeing participation) — primary cause and the instrument cascade (paraphrased). — Retrieved from: _____
2. Peru CIAA report (1997) — the static-port tape and the contradictory warnings. — Retrieved from: _____
3. Peru CIAA report (1997) — the absence of an independent cockpit reference. — Retrieved from: _____
4. Leveson, N. (2011), Engineering a Safer World (STAMP) — common-cause failure. — Retrieved from: _____
5. Dismukes, Berman & Loukopoulos (2007), The Limits of Expertise — crew performance under instrument failure. — Retrieved from: _____

CASE 97 Boeing 737 MAX / MCAS

2018 – 2019

1. U.S. House Committee on Transportation and Infrastructure, The Boeing 737 MAX Aircraft: Costs, Consequences, and Lessons from Its Design, Development, and Certification (Final Committee Report, Sept. 2020) — the MCAS omission, the 2013 minutes, the 2016 survey, and the “diminished safety” conclusion. — Retrieved from: _____
2. U.S. House report (2020) and MCAS design summary — MCAS firing on a single angle-of-attack sensor; the Lion Air 610 (189 killed) and Ethiopian 302 (157 killed) accident sequences. — Retrieved from: _____
3. U.S. House Transportation Committee report (2020) — Boeing internal communications and the certification-and-training-impact reasoning (quoted). — Retrieved from: _____
4. U.S. Department of Transportation, Office of Inspector General, correspondence and review of FAA oversight of MCAS and the angle-of-attack systems (2019–2020) — the delegated-certification process. — Retrieved from: _____
5. J. Herkert, J. Borenstein & K. Miller, “The Boeing 737 MAX: Lessons for Engineering Ethics,” Science and Engineering Ethics 26 (2020) — certification-by-omission as an engineering-ethics failure. — Retrieved from: _____
6. Federal Aviation Administration, Summary of the FAA’s Review of the Boeing 737 MAX (Return-to-Service, 2020) — the MCAS redesign and the simulator-training requirement; and the Aircraft Certification, Safety, and Accountability Act (2020). — Retrieved from: _____

CASE 98 Mars Climate Orbiter — Unit Mismatch

1999

1. NASA, Mars Climate Orbiter Mishap Investigation Board: Phase I Report (Nov. 1999) — mission, the Lockheed/JPL split, and the program context. — Retrieved from: _____
2. NASA MCO MIB Report (1999) — the pound-force-second vs newton-second mismatch (factor 4.45), the accumulated navigation error, and atmospheric destruction on 23 Sept. 1999 (root-cause statement quoted). — Retrieved from: _____
3. NASA MCO MIB Report (1999) — the missing verified interface specification, the absent end-to-end validation, and the unresolved cruise-trajectory anomalies. — Retrieved from: _____
4. N. G. Leveson, Engineering a Safer World (MIT Press, 2011) — interfaces as engineering deliverables requiring an owner and a verification step. — Retrieved from: _____
5. B. Sauser et al. (2009), retrospective analysis of the Mars Climate Orbiter and the “faster, better, cheaper” trade space. — Retrieved from: _____

CASE 100 Glass-Cockpit Transition in Light Aircraft — Technology Outran Training 2010

1. National Transportation Safety Board (2010). Introduction of Glass Cockpit Avionics into Light Aircraft, Safety Study NTSB/SS-10/01. <https://www.nts.gov/safety/safety-studies/Documents/SS1001.pdf> — the case's primary investigation document. — Retrieved from: _____
2. NTSB Safety Recommendations A-10-36 through A-10-41 (2010), issued to the FAA — the engineering response the open verdict points to. — Retrieved from: _____
3. Wiener, E. L., & Curry, R. E. (1980). Flight-deck automation: Promises and problems. *Ergonomics*, 23(10):995–1011 — the foundational literature on automation-induced failure modes that the glass-cockpit transition re-introduced at the GA scale. — Retrieved from: _____
4. Sarter, N. B., Woods, D. D., & Billings, C. E. (1997). Automation surprises. In *Handbook of Human Factors and Ergonomics* (2nd ed.) — the mode-confusion / automation-surprise literature the NTSB findings cross-reference. — Retrieved from: _____

CASE 101 Ariane 5 Flight 501 1996

1. J. L. Lions (chair), Ariane 5 Flight 501 Failure Inquiry Board Report (1996) — the data-conversion overflow (quoted). — Retrieved from: _____
2. Lions Report (1996) — the 37-second breakup and the lost payloads. — Retrieved from: _____
3. Lions Report (1996) — the 64-bit-to-16-bit conversion overflow and the simultaneous shutdown of both inertial reference systems. — Retrieved from: _____
4. Lions Report (1996) — the reused code path neither removed nor re-verified for Ariane 5's envelope. — Retrieved from: _____
5. N. Leveson, Safeware (1995) — software-reuse hazards; G. Le Lann, "An Analysis of the Ariane 5 Flight 501 Failure" (1997). — Retrieved from: _____
6. Cf. Patriot (Case 129) and Mars Climate Orbiter (Case 98). — Retrieved from: _____

CASE 102 Air France Flight 447 2009

1. Bureau d'Enquêtes et d'Analyses (BEA), Final Report on the accident on 1 June 2009 to the Airbus A330-203 registered F-GZCP operated by Air France flight AF447 (July 2012) — full report: flight, aircraft, and the known pitot-icing vulnerability with retrofit pending. — Retrieved from: _____
2. BEA, Final Report AF447 (2012) — accident sequence: autopilot/autothrust disconnection, degraded control law, sustained nose-up input, stall, and the stall-warning logic dropping out at high angle of attack. — Retrieved from: _____
3. BEA, Final Report AF447 (2012) — probable cause: airspeed inconsistency from pitot icing, inappropriate control inputs, and failure to recognize and recover from the stall. — Retrieved from: _____
4. BEA, Final Report AF447 (2012) — finding on warning-system ergonomics and stall-training conditions (quoted). — Retrieved from: _____
5. G. Palmer / E. Strickland, "Air France Flight 447 Crash Caused by a Combination of Factors," *IEEE Spectrum* (2014) — analysis of the divergence between the trained and operational envelopes. — Retrieved from: _____
6. BEA, Final Report AF447 (2012), safety recommendations — pitot-probe replacement, manual high-altitude flying, approach-to-stall and stall recovery, unreliable-airspeed procedures, and angle-of-attack indication. — Retrieved from: _____

7. European Union Aviation Safety Agency, Upset Prevention and Recovery Training (UPRT) requirements for airline pilots (phased in by 2019); ICAO, Manual on Aeroplane Upset Prevention and Recovery Training (Doc 10011). — Retrieved from: _____
8. Federal Aviation Administration, FAA-S-ACS-1, Airline Transport Pilot Practical Test Standards and the 2014 Part 121 stall-recovery training rule reflecting BEA recommendations; FAA Advisory Circular 120-109A, Stall Prevention and Recovery Training. — Retrieved from: _____

CASE 103 **Kegworth / British Midland 92**

1989

1. Air Accidents Investigation Branch, Report on the accident to Boeing 737-400 G-OBME near Kegworth, Leicestershire, on 8 January 1989, AAIB Aircraft Accident Report 4/90 (1990) — aircraft, route, and the limited conversion training; see also U.S. FAA, Lessons Learned: G-OBME. — Retrieved from: _____
2. AAIB Report 4/90 (1990) — accident sequence: left-engine fan-blade failure, shutdown of the serviceable right engine, and total power loss on approach; 47 killed, 74 seriously injured. — Retrieved from: _____
3. AAIB Report 4/90 (1990) — the older-737 bleed-air mental model misapplied to the 737-400's changed configuration. — Retrieved from: _____
4. AAIB Report 4/90 (1990) — the new electronic engine instrumentation and the readability of the vibration indicator. — Retrieved from: _____
5. AAIB Report 4/90 (1990) — the cabin-to-flight-deck communication gap (crew resource management). — Retrieved from: _____
6. AAIB Report 4/90 (1990) — the 31 safety recommendations on difference training, instrument design, CRM, and crashworthiness. — Retrieved from: _____
7. J. Reason, Human Error (Cambridge Univ. Press, 1990) — Kegworth as a case in the misapplication of a correct mental model to a changed system. — Retrieved from: _____

CASE 104 **Asiana Airlines Flight 214**

2013

1. NTSB, Aircraft Accident Report: Asiana Airlines Flight 214, NTSB/AAR-14/01 (2014) — probable cause and contributing factors (quoted). — Retrieved from: _____
2. NTSB/AAR-14/01 (2014) — the autothrottle HOLD reversion and airspeed decay to 103 kt. — Retrieved from: _____
3. NTSB/AAR-14/01 (2014) — the captain's mental model of the autothrottle. — Retrieved from: _____
4. Sarter, N. & Woods, D. (1995), "How in the world did we ever get into that mode?," Human Factors — automation surprise. — Retrieved from: _____
5. NTSB/AAR-14/01 (2014), safety recommendations on autoflight training and mode annunciation. — Retrieved from: _____

CASE 105 **Helios Airways Flight 522**

2005

1. Hellenic Air Accident Investigation & Aviation Safety Board, Aircraft Accident Report 11/2006 — probable cause and the warning misinterpretation (paraphrased). — Retrieved from: _____
2. AAIASB Report 11/2006 — the manual pressurization selector and failure to pressurize. — Retrieved from: _____
3. AAIASB Report 11/2006 — the dual-meaning warning horn and the training emphasis. — Retrieved from: _____
4. Boeing 737 Flight Crew Operations Manual — configuration and cabin-altitude warnings. — Retrieved from: _____

5. Reason, J. (1990), Human Error — the cue-ambiguity / human-error framing the case draws on. — Retrieved from: _____

CASE 106 TransAsia Airways Flight 235

2015

1. Aviation Safety Council (Taiwan), Aviation Occurrence Report: TransAsia Airways Flight GE235, final report (2016) — findings and the wrong-engine shutdown (paraphrased). — Retrieved from: _____
2. ASC final report (2016) — the autofeather event and the crash sequence. — Retrieved from: _____
3. ASC final report (2016) — non-use of the checklist and the captain's proficiency record. — Retrieved from: _____
4. AAIB (UK), British Midland Flight 92 (Kegworth) report (1990) — the antecedent wrong-engine case. — Retrieved from: _____
5. Dismukes, Berman & Loukopoulos (2007), The Limits of Expertise — memory-driven response under time pressure. — Retrieved from: _____

CASE 107 Eastern Air Lines Flight 401

1972

1. NTSB, Aircraft Accident Report: Eastern Air Lines Flight 401, NTSB-AAR-73-14 (1973) — the night approach and full flight deck. — Retrieved from: _____
2. NTSB-AAR-73-14 (1973) — the landing-gear-bulb fixation, the inadvertent autopilot disengagement, and 101 deaths of 176 aboard. — Retrieved from: _____
3. NTSB-AAR-73-14 (1973) — the crew's preoccupation with the landing-gear indication that distracted them from maintaining altitude (paraphrased). — Retrieved from: _____
4. C. D. Wickens et al., Engineering Psychology and Human Performance — attention, monitoring, and alert design. — Retrieved from: _____
5. R. Helmreich & H. Foushee (1993), aircrew-coordination research — the Crew Resource Management origins. — Retrieved from: _____
6. Eastern 401 as a formative case behind CRM (Case 117) and prioritized cockpit alerting standards. — Retrieved from: _____

CASE 108 NASA EVA-23 — Water Intrusion Inside a Spacesuit

2013

1. NASA International Space Station Program (2013), "International Space Station EVA Suit Water Intrusion High Visibility Close Call" — Mishap Investigation Board final report, December 2013. — Retrieved from: _____
2. NASA Aerospace Safety Advisory Panel (2014), Annual Report — EVA-23 follow-up actions including helmet absorption pad and snorkel. — Retrieved from: _____
3. ESA (2013), debrief and operational statement on the Parmitano EVA-23 event. — Retrieved from: _____
4. Chappell, Norcross, Abercromby et al. (2017), "Risk of Injury and Compromised Performance Due to EVA Operations," NASA Human Research Program Evidence Report — broader EVA risk context. — Retrieved from: _____

CASE 109 Colgan Air Flight 3407

2009

1. NTSB, Aircraft Accident Report: Colgan Air Flight 3407, NTSB/AAR-10/01 (2010) — probable cause (quoted). — Retrieved from: _____
2. NTSB/AAR-10/01 (2010) — the captain's training history, and the first officer's fatigue and pay. — Retrieved from: _____

- 3. NTSB/AAR-10/01 (2010) — the information-flow gap between known pilot history and the hiring decision. — Retrieved from: _____
- 4. Airline Safety and Federal Aviation Administration Extension Act of 2010, Pub. L. 111-216 — the 1,500-hour rule and the Pilot Records Database. — Retrieved from: _____
- 5. GAO-12-203, Pilot Records Database Implementation (2012); the Families of Continental Flight 3407 campaign. — Retrieved from: _____

CASE 110 **Atlas Air Flight 3591**

2019

- 1. NTSB, Aircraft Accident Report: Atlas Air Flight 3591, NTSB/AAR-20/02 (2020) — probable cause and contributing factors (paraphrased). — Retrieved from: _____
- 2. NTSB/AAR-20/02 (2020) — the inadvertent go-around activation and the dive into Trinity Bay. — Retrieved from: _____
- 3. NTSB/AAR-20/02 (2020) — the first officer’s training history and Atlas’s record-access limits. — Retrieved from: _____
- 4. FAA Pilot Records Database final rule (Federal Register, 10 June 2021; subpart B/C compliance from June 2022, full historical coverage by September 2024); Pilot Records Improvement Act of 1996. — Retrieved from: _____
- 5. GAO reports on Pilot Records Database implementation (2014–2024). — Retrieved from: _____

CASE 111 **Challenger & Columbia**

1986 / 2003

- 1. Rogers Commission, Report of the Presidential Commission on the Space Shuttle Challenger Accident (1986) — the O-ring failure, the Thiokol teleconference, and the cold-weather launch decision. — Retrieved from: _____
- 2. D. Vaughan, The Challenger Launch Decision: Risky Technology, Culture, and Deviance at NASA (Univ. of Chicago Press, 1996) — normalization of deviance; the working group’s drift of decision rules. — Retrieved from: _____
- 3. Columbia Accident Investigation Board, Report Vol. I (2003) — the foam strike, the sixteen prior missions of foam-shedding filed as “in-family,” the recurrence of the cultural pattern, and the call for an independent technical authority. — Retrieved from: _____
- 4. CAIB (2003) — “the NASA organizational culture had as much to do with this accident as the foam” (quoted); Rogers Commission (1986) and CAIB (2003) jointly on the suppressed upward flow of safety information across both accidents. — Retrieved from: _____
- 5. W. Starbuck & M. Farjoun (eds.), Organization at the Limit: Lessons from the Columbia Disaster (2005) — independent academic re-analyses of the institutional-learning gap. — Retrieved from: _____
- 6. NASA Engineering and Safety Center founding documents (2003 – present) — institutional response to the CAIB’s call for independent technical authority; Shuttle retirement (STS-135, July 2011). — Retrieved from: _____

CASE 112 **NASA Saturn V Documentation**

1972 – present

- 1. NASA Constellation Program documentation and reviews (2005–2010) — the difficulty of reproducing Saturn V capability. — Retrieved from: _____
- 2. R. E. Bilstein, Stages to Saturn: A Technological History of the Apollo/Saturn Launch Vehicles (NASA SP-4206, 1980) — the program and its workforce. — Retrieved from: _____
- 3. The documentation-vs-capability distinction (editors’ synthesis of the Saturn V record). — Retrieved from: _____

- 4. M. Polanyi, The Tacit Dimension (1966); I. Nonaka & H. Takeuchi, The Knowledge–Creating Company (1995). — Retrieved from: _____
- 5. J. S. Brown & P. Duguid, The Social Life of Information (2000) — tacit and institutional knowledge. — Retrieved from: _____

CASE 113 **Boeing Starliner**2019 – 2024

- 1. U.S. GAO, NASA Commercial Crew Program, GAO–20–121 (Jan. 2020) and GAO–19–504 (2019) — schedule slips and technical risks across the program. — Retrieved from: _____
- 2. The Starliner test history — 2019 software errors, 2021 valve scrub, and the 2024 crewed flight and ISS stay. — Retrieved from: _____
- 3. NASA reviews and reporting on the Starliner test program — issues across software, valves, propulsion, and integration testing (editors’ synthesis). — Retrieved from: _____
- 4. NASA Aerospace Safety Advisory Panel reports (2020–2024) — the institutional and generational factors. — Retrieved from: _____
- 5. The 2024–2025 return of the Starliner crew aboard a SpaceX vehicle. — Retrieved from: _____
- 6. Cf. Saturn V (Case 112); N. Augustine, Augustine’s Laws (1986). — Retrieved from: _____

CASE 114 **FAA Aging–Aircraft Program — Widespread Fatigue Damage and the Part 26 Rule**1988 – 2010s

- 1. NTSB (1989), Aircraft Accident Report AAR–89/03, Aloha Airlines Flight 243, Boeing 737–200, N73711. — Retrieved from: _____
- 2. FAA, 14 CFR Part 26, “Continued Airworthiness and Safety Improvements for Transport Category Airplanes,” Final Rule (2007); WFD/LOV Final Rule published Nov 15, 2010, effective Jan 14, 2011. — Retrieved from: _____
- 3. Airworthiness Assurance Working Group reports (1990s–2000s) — per-model structural inspection program record. — Retrieved from: _____
- 4. FAA Aging Aircraft Program documentation (1988 – present) — institutional record of the sustained regulatory work. — Retrieved from: _____
- 5. Swift, “Widespread Fatigue Damage Monitoring — Issues and Concerns,” International Conference on Aging Aircraft — technical synthesis of the WFD inspection regime. — Retrieved from: _____

CASE 115 **FAA NextGen and the ADS–B Out Transition**2003 – 2020

- 1. FAA, “Automatic Dependent Surveillance – Broadcast (ADS–B) Out Performance Requirements to Support Air Traffic Control (ATC) Service,” Final Rule, 14 CFR Part 91 (2010). — Retrieved from: _____
- 2. Vision 100 — Century of Aviation Reauthorization Act, Public Law 108–176 (2003) — NextGen program statutory basis. — Retrieved from: _____
- 3. Government Accountability Office, “NextGen Air Transportation System” report series (2010s) — sustained external audit record on schedule slippage and benefit–realization gaps. — Retrieved from: _____
- 4. Department of Transportation Office of Inspector General, NextGen audit report series — independent program review documenting cost growth and partial benefit realization. — Retrieved from: _____
- 5. FAA, NextGen Annual Reports (2010 – present) — program–self–report tier; read against the GAO and DOT IG reviews. — Retrieved from: _____

CASE 116 Eurocat ATM Pilot Modernization — Small-Tier Verification as the Gateway to Big-Tier Change 2005

1. DelaPeyronnie, Newcomb, Morillo, Trimech, Nguyen, & Purtil (2010), "Modernization of the Eurocat Air Traffic Management System (EATMS)," in Information Systems Transformation: Architecture-Driven Modernization Case Studies (Elsevier / Morgan Kaufmann), Chapter 5. — Retrieved from: _____
2. Ulrich & Newcomb (eds., 2010), Information Systems Transformation — the host volume on architecture-driven modernization patterns. — Retrieved from: _____
3. Brodie & Stonebraker (1995), Migrating Legacy Systems — the framing reference on small-step legacy modernization. — Retrieved from: _____
4. Bisbal et al. (1999), "Legacy Information Systems: Issues and Directions," IEEE Software 16(5):103–111 — peer-reviewed framing companion. — Retrieved from: _____

CASE 117 Crew Resource Management & CAST 1981 – present

1. FAA Advisory Circular 120–51E, Crew Resource Management Training — CRM protocols and authority-gradient training. — Retrieved from: _____
2. Helmreich, Merritt & Wilhelm (1999), "The Evolution of Crew Resource Management Training in Commercial Aviation," International Journal of Aviation Psychology. — Retrieved from: _____
3. Spanish CIAIAC / ALPA reports on the 1977 Tenerife collision — the overridden crew challenge. — Retrieved from: _____
4. CAST/FAA Safety Enhancement reports (2016, 2018) — the closed-loop data process and the 83% reduction. — Retrieved from: _____
5. Collier Trophy citation (2008); Kanki, Helmreich & Anca (2010), Crew Resource Management. — Retrieved from: _____

CASE 118 Korean Air Safety Transformation 2000 – present

1. NTSB, Aircraft Accident Report: Korean Air Flight 801, Guam (2000) — authority gradients as a root cause. — Retrieved from: _____
2. Air Transport World Phoenix Award documentation (2006) — recognition of the turnaround. — Retrieved from: _____
3. Gladwell, M. (2008), Outliers — the Korean Air chapter on power distance and the English-language change. — Retrieved from: _____
4. Helmreich, Wilhelm, Klinec & Merritt (2001) — national culture and CRM adaptation. — Retrieved from: _____
5. Korean Air corporate safety reports — the post-2000 accident-free record. — Retrieved from: _____

CASE 119 Aviation Safety Reporting System (ASRS) 1976 – present

1. NASA ASRS Program documentation and annual reports — system design, immunity provision, and report volume. — Retrieved from: _____
2. Reason, J. (1997), Managing the Risks of Organizational Accidents — non-punitive reporting as a model (paraphrased). — Retrieved from: _____
3. NASA ASRS technical reports (Connell et al.) — patterns first surfaced via ASRS data. — Retrieved from: _____

4. Dekker, S. (2012), Just Culture — the cultural commitment to non-punitive use. — Retrieved from: _____
5. CHIRP and patient-safety reporting-system documentation — cross-domain emulation. — Retrieved from: _____

CASE 120 EGPWS / TAWS — Closing the CFIT Category in Commercial Aviation 1996 – 2002

1. Bateman, C. D. (1999), "The Introduction of Enhanced Ground Proximity Warning Systems (EGPWS) into Civil Aviation Operations Around the World," in Proceedings of the 11th Annual European Aviation Safety Seminar, Flight Safety Foundation, pp. 259–274 — developer history. — Retrieved from: _____
2. Federal Aviation Administration (2000), 14 CFR §§ 91.223, 121.354, 135.154 — Terrain Awareness and Warning System (TAWS) equipage requirement. — Retrieved from: _____
3. Flight Safety Foundation (1998 – 2000), CFIT / ALAR Task Force reports — operational and outcome analyses motivating mandate. — Retrieved from: _____
4. NTSB (1996), Aircraft Accident Report AAR-96-05, American Airlines 965, Cali, Colombia, December 20 1995. — Retrieved from: _____
5. Royal Commission to Inquire into the Crash on Mount Erebus, Antarctica of a DC10 Aircraft Operated by Air New Zealand Limited (1981), final report (Mahon report). — Retrieved from: _____

CASE 121 TCAS — Coordinated Collision Avoidance and the Überlingen Lesson 1981 – 2008 (TCAS II Version 7.1)

1. RTCA (2008), DO-185B "Minimum Operational Performance Standards for Traffic Alert and Collision Avoidance System II (TCAS II)" — Version 7.1 with reversal logic. — Retrieved from: _____
2. Bundesstelle für Flugunfalluntersuchung (BFU) (2004), AX001-1-2/02 — Investigation Report on the mid-air collision on 1 July 2002 near Überlingen. — Retrieved from: _____
3. Eurocontrol (2008 – 2011), ACAS II Bulletin and Programme for the Harmonised Implementation of Satellite Navigation — V7.1 mandate timing in Europe. — Retrieved from: _____
4. Federal Aviation Administration, 14 CFR § 121.356 — TCAS II equipage requirement. — Retrieved from: _____
5. NTSB / FAA TCAS historical safety studies (1990 – 2010) — outcome record context. — Retrieved from: _____

CASE 122 Singapore Airlines Safety Transformation 1980s – present

1. Aviation Safety Council (Taiwan), SQ006 Accident Investigation Final Report (2002) — the accident and the airline's response. — Retrieved from: _____
2. Singapore Airlines safety reports (multiple years) — training, fleet, and reporting-culture investment. — Retrieved from: _____
3. IATA Operational Safety Audit (IOSA) documentation — investment ahead of regulatory minimums (paraphrased). — Retrieved from: _____
4. Helmreich, Wilhelm, Klinec & Merritt (2001) — national culture and CRM adaptation. — Retrieved from: _____
5. Weick & Sutcliffe (2007), Managing the Unexpected — sustained high-reliability investment. — Retrieved from: _____

CASE 123

FSF CFIT and ALAR Task Forces — Industry-Level Institution Building After a Spike

1992 – 2000s

1. Flight Safety Foundation, “Killers in Aviation: FSF Task Force Presents Facts About Approach-and-landing and Controlled-flight-into-terrain Accidents,” Flight Safety Digest (1998–1999). — Retrieved from: _____
2. Flight Safety Foundation, ALAR Tool Kit (1998) — distributed multi-element package of briefing notes, training aids, and risk-assessment instruments. — Retrieved from: _____
3. Khatwa & Helmreich, “Analysis of critical factors during approach and landing in accidents and normal flight,” Flight Safety Digest (1998) — the analytical basis of the ALAR Task Force scope. — Retrieved from: _____
4. FAA, Terrain Awareness and Warning System (TAWS) Final Rule, 14 CFR Part 121 (2000, effective 2002). — Retrieved from: _____
5. ICAO Annex 6, TAWS requirement (effective 2007) — the international consolidation. — Retrieved from: _____

CASE 124

USS Fitzgerald & USS John S. McCain

2017

1. T. C. Miller, M. Faturechi & R. Rotella, “Years of Warnings, Then Death and Disaster,” ProPublica (2019) — the 2003 SWOS-in-a-Box shift. — Retrieved from: _____
2. GAO, Navy Readiness, GAO-17-809T (Sept. 2017) — 37% of Japan-based warfare certifications expired by June 2017. — Retrieved from: _____
3. NTSB, Collision between USS Fitzgerald and MV ACX Crystal, NTSB/MAR-20/02 (2020), DCA17PM018. — Retrieved from: _____
4. NTSB, Collision between USS John S. McCain and Tanker Alnic MC, NTSB/MAR-19/01 (2019), DCA17PM029. — Retrieved from: _____
5. U.S. Navy, Comprehensive Review of Recent Surface Force Incidents (Nov. 2017) — both collisions judged avoidable. — Retrieved from: _____
6. NTSB/MAR-19/01 (2019) — probable cause: insufficient training and inadequate bridge procedures from a lack of Navy oversight. — Retrieved from: _____
7. R. Rotella et al., “The Navy Installed Touch-Screen Steering Systems to Save Money,” ProPublica (2019). — Retrieved from: _____
8. U.S. Navy, Strategic Readiness Review (Dec. 2017) — risks that “accumulated over time and did so insidiously.” — Retrieved from: _____
9. U.S. Navy corrective actions (2017); NTSB/MAR-19/01 recommendations; USNI News (2017–2020). — Retrieved from: _____
10. Surface Warfare Officers School Command, Basic Division Officer Course curriculum and the post-2017 return to in-person instruction; Naval Surface Group Western Pacific stand-up (2019) as the forward-readiness certification authority. — Retrieved from: _____
11. NTSB/MAR-19/01 (2019) — watchstander fatigue findings, including average sleep hours and the touch-screen helm misdiagnosis sequence. — Retrieved from: _____

CASE 126

F-35 Sustainment & Maintainer Shortage

ongoing

1. U.S. Government Accountability Office, F-35 Aircraft: DOD and the Military Services Need to Reassess the Future Sustainment Strategy, GAO-23-105341 (Sept. 2023) — 2,500 planned aircraft and a lifecycle cost exceeding \$1.7 trillion, \$1.3 trillion of it in operating and support. — Retrieved from: _____

2. GAO-23-105341 (2023) — 55% fleet mission-capable rate (March 2023); over 10,000 components awaiting repair; 72-day average depot turnaround; depot stand-up behind schedule. — Retrieved from: _____
3. GAO-23-105341 (2023) — sustainment shortfall traced to maintainer shortages, lack of military access to technical data, and contractor dependency. — Retrieved from: _____
4. GAO, F-35 Sustainment: DOD Faces Several Uncertainties..., GAO-22-105995 (2022), and the broader GAO F-35 series — the recurring, year-over-year diagnosis. — Retrieved from: _____
5. GAO-23-105341 (2023) — recommendation that DOD reassess the future sustainment strategy. — Retrieved from: _____
6. GAO, F-35 Sustainment: Costs Continue to Rise While Planned Use and Availability Have Decreased, GAO-24-106703 (2024) — costs rising while readiness stays below goal. — Retrieved from: _____

CASE 127 Military Fratricide — Desert Storm to Afghanistan

1991 – 2014

1. C. R. Shrader, Amicide: The Problem of Friendly Fire in Modern War (U.S. Army Combat Studies Institute, 1982) — the under-2% estimate from 269 incidents, a baseline later challenged as too low (with estimates nearer 10–15%). — Retrieved from: _____
2. “Friendly Fire: Facts, Myths and Misperceptions,” U.S. Naval Institute Proceedings (June 1994) — Desert Storm: 35 of 146 KIA (24%) and 72 of 467 wounded (15%) by friendly fire; critique of the 2% “historical norm.” — Retrieved from: _____
3. Khafji and Warrior fratricide incidents (Feb. 1991) — see USNI Proceedings (1994) and Time, “They Didn’t Have to Die”. (Per-incident casualty figures vary across sources; see AUDIT.) — Retrieved from: _____
4. U.S. GAO, Operation Desert Storm fratricide investigations — Apache incident (OSI-93-4) and Army fratricide investigation (OSI-95-10) — causes and the Army’s reviews. — Retrieved from: _____
5. Post-war combat-identification reviews — leading causes (situational awareness, fire-control measures, combat ID) and the absence of a comprehensive incident database. (Synthesized from secondary analyses; see AUDIT.) — Retrieved from: _____
6. Later-conflict recurrences (2001 coordinate-reset strike; 2014 B-1 infrared-strobe strike) and S. Snook, Friendly Fire (Princeton Univ. Press, 2000) on the 1994 Black Hawk shootdown — the systems-integration archetype. — Retrieved from: _____
7. “IFF Update: Stalled Again,” U.S. Naval Institute Proceedings (June 1994) — the combat-identification and identification-friend-or-foe programs pursued after Desert Storm, and how slowly they matured. — Retrieved from: _____

CASE 128 Operation Eagle Claw

1980

1. Holloway Special Operations Review Group, Rescue Mission Report (1980) — the ad-hoc assembly and equipment choices (paraphrased). — Retrieved from: _____
2. Holloway Commission (1980) — the Desert One sequence and the helicopter-C-130 collision. — Retrieved from: _____
3. Goldwater-Nichols Department of Defense Reorganization Act of 1986, Pub. L. 99-433. — Retrieved from: _____
4. Nunn-Cohen Amendment (1986), establishing U.S. Special Operations Command (1987). — Retrieved from: _____
5. Locher, J. (2002), Victory on the Potomac — the reform arc from Desert One to Goldwater-Nichols. — Retrieved from: _____

CASE 129 Patriot Missile / Dhahran

1991

1. U.S. General Accounting Office, Patriot Missile Defense: Software Problem Led to System Failure at Dhahran, Saudi Arabia, GAO/IMTEC-92-26 (1992) — the design-context mismatch and continuous-operation use. — Retrieved from: _____
2. R. Skeel, "Roundoff Error and the Patriot Missile," SIAM News 25(4): 11 (1992) — the 24-bit fixed-point time truncation and the 0.34-second range-gate drift after 100 hours. — Retrieved from: _____
3. GAO/IMTEC-92-26 (1992) — the prior Israeli warning, the software patch arriving in Dhahran on 26 February (the day after the strike), and the 28 killed. — Retrieved from: _____
4. GAO/IMTEC-92-26 (1992) — the vague "very long run time" advisory and the Army's presumption that batteries would not be run continuously for such periods (quoted). — Retrieved from: _____
5. M. Barr / Barr Group, "An Update on the Patriot Missile Software Problem" — an engineering post-mortem of the accumulated-truncation defect. — Retrieved from: _____
6. GAO/IMTEC-92-26 (1992) and Skeel (1992) — the design assumption that failed to travel across the change in operational context, and the case's teaching legacy. — Retrieved from: _____

CASE 130 USS Vincennes & Iran Air Flight 655

1988

1. Rear Adm. W. Fogarty, Formal Investigation into the Circumstances Surrounding the Downing of Iran Air Flight 655 (US Navy, August 1988) — the engagement, the IFF mode-confusion findings, and the 290 deaths. — Retrieved from: _____
2. Fogarty report (1988) — the Aegis SPY-1A radar functioned; the aircraft was ascending while the crew perceived a descent; "scenario fulfillment" as the psychological mechanism. — Retrieved from: _____
3. Fogarty report (1988) — "human error under extreme stress," confirmation bias, and "unconscious distortion of data" (quoted); shared-error finding across CIC operators. — Retrieved from: _____
4. J. Barry & R. Charles, "Sea of Lies," Newsweek (July 13, 1992) — contemporaneous reinvestigation of the operational record, including the disputed account of Vincennes' position relative to Iranian territorial waters. — Retrieved from: _____
5. US Naval Institute Proceedings retrospective on the Vincennes incident (2018) — the human-AI-teaming reframing and the standing case-study role. — Retrieved from: _____
6. M. L. Cummings, "Human Supervisory Control of Weapon Systems" (MIT) — interface design and automation under time pressure as the engineering frame for the case. — Retrieved from: _____
7. G. Klein, Sources of Power (1998); M. Endsley, "Toward a Theory of Situation Awareness" (1995) — the naturalistic-decision-making and situation-awareness literatures that treat Vincennes as the canonical worked example. — Retrieved from: _____

CASE 131 Stanislav Petrov / 1983 False Alert

1983

1. D. Hoffman, The Dead Hand: The Cold War Arms Race and Its Dangerous Legacy (2009) — the Oko system and Soviet launch-on-warning posture. — Retrieved from: _____
2. Accounts of the 26 Sept. 1983 incident (Hoffman; Petrov interviews) — the five-missile alert and Petrov's assessment. — Retrieved from: _____
3. Investigations of the Oko incident (incl. V. Aksenov, 2004) — sunlight-on-clouds satellite geometry as the false-positive cause. — Retrieved from: _____
4. The subsequent Oko algorithm modification and the incident's use in early-warning training. — Retrieved from: _____

5. S. Petrov, interview, The Washington Post (1999) — “a funny feeling in my gut” (quoted). — Retrieved from: _____

6. S. Sagan, The Limits of Safety (1993); M. L. Cummings (2017) — automation in critical decisions and human-in-the-loop. — Retrieved from: _____

CASE 132 **F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap** 2008 – 2012

1. USAF Scientific Advisory Board (2012), “Aircraft Oxygen Generation Study” — final report on F-22 physiological-event investigation. — Retrieved from: _____

2. NASA Engineering and Safety Center (2012), report contributing to the F-22 hypoxia review. — Retrieved from: _____

3. USAF Accident Investigation Board (2011), Capt. Jeffrey Haney F-22 accident report (Nov 2010). — Retrieved from: _____

4. GAO-12-789 (2012), “Actions Needed to Establish Effective Oversight of F-22 Pilot Physiological Issues” — congressional record. — Retrieved from: _____

CASE 133 **Mark 14 Torpedo Failures** 1941 – 1943

1. Blair, C. (1975), Silent Victory: The U.S. Submarine War Against Japan — operator reports and the Bureau’s resistance (paraphrased). — Retrieved from: _____

2. Rowland, B. & Boyd, W. (1953), U.S. Navy Bureau of Ordnance in World War II — testing policy and the three defects. — Retrieved from: _____

3. Blair (1975) — the USS Tinosa’s July 1943 attack (eleven consecutive duds on a stopped target) and Lockwood’s fleet-level testing. — Retrieved from: _____

4. Naval History and Heritage Command, Mark 14 torpedo files — depth error, Mark 6 magnetic exploder, contact-pin failure. — Retrieved from: _____

5. Edmondson, A. (2018), The Fearless Organization — suppression of field reports as an organizational-learning failure. — Retrieved from: _____

CASE 134 **V-22 Osprey** 1991 – present

1. Compiled V-22 accident record — 17 hull losses and 65 fatalities since 1991, including the 2000 Marana test crash (19 Marines). — Retrieved from: _____

2. U.S. Air Force Accident Investigation Board findings, via USNI News (Aug. 2024) — the 29 Nov. 2023 Yakushima CV-22B crash: transmission-gear cracks (X-53 inclusions) and continued flight despite warnings. — Retrieved from: _____

3. U.S. GAO, Osprey Aircraft: Additional Oversight and Information Sharing Would Improve Safety Efforts, GAO-26-108905 (Dec. 2025) — the joint program office’s failure to assess and address risks. — Retrieved from: _____

4. GAO-26-108905 (2025) — serious-mishap rates exceeding comparable Navy/Air Force fleets (FY2015–FY2024); the 2006 gearbox flaw evaluated only in 2024; fixes into the 2030s. — Retrieved from: _____

5. NAVAIR independent review of the V-22 (Dec. 2025) — materiel and cross-service-coordination factors and unresolved catastrophic parts issues. — Retrieved from: _____

6. USNI News, V-22 program coverage (2024–2025). — Retrieved from: _____

CASE 135 9/11 Intelligence Sharing Failures

1996 – 2001

1. National Commission on Terrorist Attacks Upon the United States, The 9/11 Commission Report (2004) — the quoted “failure of imagination” and the specific sharing failures. — Retrieved from: _____
2. 9/11 Commission Report (2004) — the Kuala Lumpur tracking and the Phoenix/Minneapolis flagging. — Retrieved from: _____
3. Joint Inquiry into Intelligence Community Activities Before and After September 11, 2001 (2002) — cross-agency information-sharing failures. — Retrieved from: _____
4. Zegart, A. (2007), Spying Blind — structural-organizational analysis of the failures. — Retrieved from: _____
5. Intelligence Reform and Terrorism Prevention Act of 2004 — creation of the ODNI and NCTC. — Retrieved from: _____

CASE 137 U.S. Nuclear Navy / Rickover Training Model

1954 – present

1. Polmar, N. & Allen, T. (2007), Rickover: Father of the Nuclear Navy — the program and Rickover’s philosophy (paraphrased). — Retrieved from: _____
2. Naval Nuclear Propulsion Program documentation (NRC/DOE) — qualification standards and the accident record. — Retrieved from: _____
3. Admiral Hyman G. Rickover, “Doing a Job” (Columbia University commencement address, 1982) — “people, not organizations... get things done” (quoted). — Retrieved from: _____
4. GAO-21-168, comparison of nuclear and surface Navy training — the internal contrast. — Retrieved from: _____
5. Duncan, F. (1990), Rickover and the Nuclear Navy — the qualification culture. — Retrieved from: _____

CASE 138 MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard

2020 revision (series since 1968)

1. Department of Defense (2020), MIL-STD-1472H “Department of Defense Design Criteria Standard: Human Engineering,” 15 September 2020 — replaces MIL-STD-1472G (2012). — Retrieved from: _____
2. Department of Defense (2012), MIL-STD-1472G — the prior revision; revision history documents the 1968 origin and intermediate letters. — Retrieved from: _____
3. Chapanis, A. (1965), “Man-Machine Engineering” — foundational text for the discipline the standard codifies. — Retrieved from: _____
4. Fitts, P. M., & Jones, R. E. (1947), “Analysis of factors contributing to 460 ‘pilot error’ experiences in operating aircraft controls” — origin of designed-error analysis. — Retrieved from: _____

CASE 139 GIFT and the Adoption Gap

2012 – present

1. K. VanLehn, “The Relative Effectiveness of Human Tutoring, Intelligent Tutoring Systems, and Other Tutoring Systems,” Educational Psychologist 46(4) (2011) — tutoring effectiveness. — Retrieved from: _____
2. GIFT Project, gifttutoring.org — the framework, releases, and development under ARL / DEVCOM. — Retrieved from: _____
3. Editors’ synthesis of the GIFT adoption record — active development but limited ubiquitous fielding (quoted). — Retrieved from: _____

4. R. Sottolare, A. Graesser, X. Hu & H. Holden (eds.), Design Recommendations for Intelligent Tutoring Systems (GIFTSym volumes); IJAIED Special Issue on GIFT (2017). — Retrieved from: _____
5. J. Goodell & J. Kolodner, Learning Engineering Toolkit (2022) — adoption as an engineering problem. — Retrieved from: _____

CASE 140 Navy Surface Warfare Readiness Reform

2018 – present

1. GAO-21-168 (and GAO-20-154), Navy readiness reform — the lack of systematic outcome evaluation (paraphrased). — Retrieved from: _____
2. Readiness Reform Oversight Council, One-Year Report (2019) — restored training, assessments, and gates. — Retrieved from: _____
3. Navy and NTSB reports on the Fitzgerald and McCain collisions (2017–2019) — the training-degradation antecedent. — Retrieved from: _____
4. SWOS Norfolk and San Diego Mariner Skills Training Center documentation — simulators and curriculum. — Retrieved from: _____
5. USNI News reform coverage (2020, 2022) — Ready-for-Sea Assessments and sidelined ships. — Retrieved from: _____

CASE 141 DARPA Digital Tutor — Compressing Years of IT Expertise into 16 Weeks

2009 – 2014

1. Fletcher, J. D., & Morrison, J. E. (2014). DARPA Digital Tutor: Assessment Data. IDA Document D-4686. <https://apps.dtic.mil/sti/tr/pdf/AD1002362.pdf> — independent evaluation that the case rests on. — Retrieved from: _____
2. Defense Advanced Research Projects Agency, Digital Tutor program documentation — program description and design rationale. — Retrieved from: _____
3. Fletcher, J. D. (2009). From behaviorism to constructivism: a philosophical journey from drill and practice to situated learning. — methodological grounding for the Digital Tutor's tutorial discipline. — Retrieved from: _____
4. Anderson, J. R., Corbett, A. T., Koedinger, K. R., & Pelletier, R. (1995). Cognitive tutors: Lessons learned. *Journal of the Learning Sciences*, 4(2):167–207. doi:10.1207/s15327809jls0402_2 — the broader intelligent-tutoring evidence base the Digital Tutor program sits within. — Retrieved from: _____

CASE 142 xAPI / Total Learning Architecture — Interoperability Gap

ongoing

1. Advanced Distributed Learning Initiative, Total Learning Architecture documentation — the cross-boundary vision. — Retrieved from: _____
2. xAPI specification (xapi.com) — the technical standard and reference implementations. — Retrieved from: _____
3. IEEE ICICLE proceedings on TLA adoption challenges — the persistence of siloed implementations. — Retrieved from: _____
4. B. Saxberg, learning-engineering infrastructure essays; IEEE ICICLE LEBok chapters on data and analytics. — Retrieved from: _____
5. Cf. inBloom (Case 53) — technology in advance of governance. — Retrieved from: _____

CASE 143 Knight Capital Trading Loss

2012

1. U.S. Securities and Exchange Commission, Order Instituting Administrative and Cease-and-Desist Proceedings, In re Knight Capital Americas LLC (2013) — the firm, the Retail Liquidity Program launch, and the deployment. — Retrieved from: _____

2. SEC Order (2013) and Knight Capital 8-K filing (2012) — the missed eighth server, the reactivation of the dormant “Power Peg” code, the \$440 million loss in 45 minutes, and the near-collapse. — Retrieved from: _____
3. SEC Order (2013) — the absence of a second-technician deployment verification, the lack of an automated code-consistency check, inadequate order controls, and the \$12 million penalty (quoted). — Retrieved from: _____
4. B. Beyer, C. Jones, J. Petoff & N. R. Murphy (eds.), *Site Reliability Engineering* (O’Reilly, 2016) — deployment verification, dead-code removal, and automated safeguards as engineering deliverables. — Retrieved from: _____
5. SEC Market Access Rule (Rule 15c3-5) and subsequent automated-controls guidance — the regulatory response on pre-trade risk and market-access controls. — Retrieved from: _____
6. D. Seven, “Knightmare: A DevOps Cautionary Tale” — a widely cited engineering post-mortem on the deployment process, the orphaned eighth server, and the reused feature flag. — Retrieved from: _____

CASE 144 Kodak Digital Camera Stagnation

1975–2012

1. S. Sasson, “We Had No Idea,” *IEEE Spectrum* (16 October 2007) — first-hand account of the 1975 prototype, the demonstration, and the internal reception. — Retrieved from: _____
2. H. C. Lucas Jr. & J. M. Goh, “Disruptive technology: How Kodak missed the digital photography revolution,” *Journal of Strategic Information Systems* 18(1):46–55 (March 2009). — Retrieved from: _____
3. Eastman Kodak Company, Voluntary Petition for Chapter 11 Reorganization, U.S. Bankruptcy Court, Southern District of New York, Case No. 12-10202 (19 January 2012). — Retrieved from: _____
4. A. R. Sorkin & M. J. de la Merced, “Eastman Kodak Files for Bankruptcy,” *The New York Times DealBook* (19 January 2012). — Retrieved from: _____
5. M. Spector & D. Mattioli, “Kodak Teeters on the Brink,” *The Wall Street Journal* (5 January 2012); follow-up coverage of the patent-sale process in *WSJ* and *Reuters* (December 2012). — Retrieved from: _____

CASE 146 Northeast Blackout

2003

1. U.S.–Canada Power System Outage Task Force, Final Report on the August 14, 2003 Blackout in the United States and Canada (2004) — the tree contact, the silent alarm, and the cascade. — Retrieved from: _____
2. Task Force (2004) — 50 million people affected across eight states and Ontario; the minute-by-minute sequence. — Retrieved from: _____
3. Task Force (2004) — FirstEnergy “did not have adequate situational awareness,” plus the vegetation-management and reliability-coordinator findings (quoted). — Retrieved from: _____
4. FERC Order No. 693, Mandatory Reliability Standards for the Bulk-Power System (2007) — enforceable standards. — Retrieved from: _____
5. North American Electric Reliability Council reports (2004) and the creation of the Electric Reliability Organization. — Retrieved from: _____
6. M. R. Endsley (1995), situation-awareness theory — the human-factors frame for silent-automation failure. — Retrieved from: _____

CASE 147 Takata Airbag Inflators

2008 – 2023

1. U.S. National Highway Traffic Safety Administration, Takata air-bag inflator recall coordination materials — the ammonium-nitrate-without-desiccant design and the propellant-degradation rupture mechanism. — Retrieved from: _____
2. NHTSA recall record and investigative reporting (*Reuters*, *New York Times*) — 100M+ inflators across 19 automakers; the largest automotive recall in history; deaths and injuries from ruptures. — Retrieved from: _____

3. U.S. Department of Justice, settlement and guilty plea with Takata Corporation (2017) — wire fraud, \$1B in fine and restitution, and Takata's subsequent bankruptcy. — Retrieved from: _____
4. U.S. DOJ (2017) and Takata internal documents released in litigation — the sustained misrepresentation of inflator test data to automakers and regulators. — Retrieved from: _____
5. NHTSA Takata recall status reporting — the years-long completion of replacements and the continuing risk in unrepaired vehicles. — Retrieved from: _____
6. NHTSA Coordinated Remedy Program for the Takata recalls — the regulator's move to actively prioritize and manage replacement across nineteen automakers rather than leave pace to each manufacturer. — Retrieved from: _____

CASE 148 Wells Fargo Fake Accounts

2011 – 2016

1. Consumer Financial Protection Bureau, Consent Order against Wells Fargo (2016) — the unauthorized accounts and penalties. — Retrieved from: _____
2. Office of the Comptroller of the Currency, Consent Order AA-EC-2016-66 (2016) — unsafe or unsound sales practices tied to the incentive-compensation structure (paraphrased). — Retrieved from: _____
3. Independent Directors of Wells Fargo, Sales Practices Investigation Report (2017) — how the sales-target architecture drove the conduct. — Retrieved from: _____
4. Enforcement record: 3.5 million accounts, \$3 billion in penalties, and the CEO's resignation. — Retrieved from: _____
5. Federal Reserve asset cap on Wells Fargo (2018) — the structural growth constraint. — Retrieved from: _____
6. A. C. Edmondson, The Fearless Organization (2018); incentive-design and corporate-governance literature. — Retrieved from: _____

CASE 149 Volkswagen Dieselgate

2015

1. U.S. EPA, Notice of Violation to Volkswagen (2015) — the defeat device and emissions exceedances. — Retrieved from: _____
2. West Virginia University / ICCT real-world diesel-emissions study (2014) — the discovery comparing road to lab. — Retrieved from: _____
3. U.S. Department of Justice Plea Agreement with Volkswagen AG (January 2017) — the institutional decision, USD 4.3 billion criminal-and-civil settlement, and convictions (quoted). — Retrieved from: _____
4. Volkswagen internal documents released through litigation — authorization of the defeat device within the engineering hierarchy. — Retrieved from: _____
5. EU Real Driving Emissions (RDE) testing regulation — the post-Dieselgate measurement reform. — Retrieved from: _____
6. J. Ewing, Faster, Higher, Farther: The Volkswagen Scandal (2017); cf. Takata (Case 147). — Retrieved from: _____

CASE 151 GM Ignition Switch

2002 – 2014

1. A. R. Valukas, Report to the Board of Directors of General Motors Company Regarding Ignition Switch Recalls (Jenner & Block, 2014) — the 2002 identification of the defect and the switch-torque mechanism. — Retrieved from: _____
2. U.S. NHTSA investigation of the GM ignition switch (2014) and the GM recall record — 2.6 million vehicles; 124 deaths via the GM compensation fund. — Retrieved from: _____

3. Valukas Report (2014) — the unchanged part number, the “GM nod” and “GM salute,” and the failure to use established escalation processes (quoted). — Retrieved from: _____
4. U.S. Department of Justice, deferred-prosecution agreement with General Motors (2015) — \$900 million forfeiture for concealing the defect; total penalties/settlements exceeding \$2.6 billion. — Retrieved from: _____
5. Valukas Report (2014) and A. C. Edmondson, *The Fearless Organization* (Wiley, 2018) — the part number as an organizational signal, and the suppression of upward safety information. — Retrieved from: _____
6. U.S. NHTSA consent order with General Motors (2014, \$35M civil penalty) and the GM ignition-switch victims’ compensation program (K. Feinberg, administrator) — the regulatory penalty and the basis for the 124-death figure. — Retrieved from: _____

CASE 152 TSB Bank IT Migration

2018

1. Slaughter and May, Independent Review of the TSB Migration (2019) — the single-weekend cutover and the testing failures. — Retrieved from: _____
2. Slaughter and May (2019) and FCA materials — 1.9 million customers locked out, £330M+ in costs, and the CEO’s resignation. — Retrieved from: _____
3. Slaughter and May (2019) — inadequate load testing and an unchallenged readiness certification. — Retrieved from: _____
4. Financial Conduct Authority, Final Notice on TSB Bank (2022) — the regulatory penalty and proceeding against technical advice. — Retrieved from: _____
5. House of Commons Treasury Committee, report on the TSB IT migration (2018). — Retrieved from: _____
6. Cf. Healthcare.gov (Case 180) and the migration-safety literature. — Retrieved from: _____

CASE 153 Equifax Data Breach

2017

1. U.S. Senate Permanent Subcommittee on Investigations, *How Equifax Neglected Cybersecurity and Suffered a Devastating Data Breach* (2019) — the unpatched Apache Struts vulnerability. — Retrieved from: _____
2. The breach record — 147 million affected, exploitation from May 2017, public disclosure in September 2017. — Retrieved from: _____
3. Senate PSI (2019) — “Equifax lacked a comprehensive IT asset inventory” (quoted). — Retrieved from: _____
4. U.S. FTC / CFPB / state settlement (\$700 million, 2019) and the CEO’s resignation. — Retrieved from: _____
5. U.S. GAO, *Actions Taken by Equifax and Federal Agencies in Response to the 2017 Breach*, GAO-18-559 (2018). — Retrieved from: _____
6. Apache Struts CVE-2017-5638 advisory. — Retrieved from: _____

CASE 154 INL Turbine-Control Upgrade — Low-Burden Cutover as a Human-Factors Finding

2014

1. Ronald L. Boring (2014), “Human Factors Design, Verification, and Validation for Two Types of Control Room Upgrades at a Nuclear Power Plant,” *Proceedings of the Human Factors and Ergonomics Society Annual Meeting (HFES 2014)*, pp. 2295–2299 (ResearchGate publication 271728006). — Retrieved from: _____
2. Idaho National Laboratory, *Light Water Reactor Sustainability Program reports on control-room modernization* — series available via OSTI. — Retrieved from: _____

3. Nuclear Regulatory Commission (NUREG-0711), “Human Factors Engineering Program Review Model” — the regulatory framework the V&V deliverables are produced against. — Retrieved from: _____
4. O’Hara et al. (2008), “Human Factors Considerations with Respect to Emerging Technology in Nuclear Power Plants,” NUREG/CR-6947 — peer-adjacent framing. — Retrieved from: _____

CASE 155 Toyota Production System / Andon Cord

1950s – present

1. Liker, J. (2020), The Toyota Way (2nd ed.) — the andon cord, the cultural pairing, and the American imitation (paraphrased). — Retrieved from: _____
2. Spear, S. & Bowen, H. (1999), “Decoding the DNA of the Toyota Production System,” HBR — the response protocol and embedded learning. — Retrieved from: _____
3. Shingo, S. (1989), A Study of the Toyota Production System — the technical mechanism. — Retrieved from: _____
4. Rother, M. (2009), Toyota Kata — the routines that sustain the practice. — Retrieved from: _____
5. Womack & Jones (1996), Lean Thinking — diffusion and the limits of surface imitation. — Retrieved from: _____

CASE 156 INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet

2010 – present

1. Idaho National Laboratory, Light Water Reactor Sustainability (LWRS) program annual reports (2010 – present) — primary research-and-pilot record. — Retrieved from: _____
2. Nuclear Regulatory Commission, Regulatory Guide 1.180, “Guidelines for Evaluating Electromagnetic and Radio-Frequency Interference in Safety-Related Instrumentation and Control Systems.” — Retrieved from: _____
3. Nuclear Regulatory Commission, Branch Technical Position 7-19, “Guidance for Evaluation of Diversity and Defense-in-Depth in Digital Computer-Based Instrumentation and Control Systems.” — Retrieved from: _____
4. O’Hara, Higgins, Brown, Fink, Persensky, Lewis, Kramer, & Szabo (2008), “Human Factors Considerations with Respect to Emerging Technology in Nuclear Power Plants,” NUREG/CR-6947 — foundational human-factors backdrop. — Retrieved from: _____
5. Electric Power Research Institute, control-room modernization technical reports — industry-side sustainment record. — Retrieved from: _____

CASE 157 AI-Augmented Coding Tools

2021 – present

1. Peng et al. (2023), “The Impact of AI on Developer Productivity” — short-term productivity gains. — Retrieved from: _____
2. Pearce et al. (2022), “Asleep at the Keyboard? Assessing the Security of GitHub Copilot’s Code Contributions.” — Retrieved from: _____
3. Sandoval et al. (2023), “Lost at C,” USENIX Security — AI assistance did not significantly increase critical security-bug rates. — Retrieved from: _____
4. Dell’Acqua et al. (2023), “Navigating the Jagged Technological Frontier” (HBS / BCG) — professional LLM use. — Retrieved from: _____
5. L. Bainbridge, “Ironies of Automation,” Automatica 19(6) (1983) — the classic deskilling and over-reliance problem, applied to AI-augmented work. — Retrieved from: _____

CASE 205 **The Discipline We Build Next**

ongoing

1. LDT / LENS program statement, Johns Hopkins School of Education — the program and its commitments. — Retrieved from: _____

2. Goodell & Kolodner (2022), Learning Engineering Toolkit — the discipline’s methods. — Retrieved from: _____

3. The preceding cases of this volume — the cumulative record of cost and of possibility. — Retrieved from: _____

4. INPO (Case 175) and CRM/CAST (Case 117) — institution-building precedents (1979, 1981). — Retrieved from: _____

5. IEEE ICICLE Learning Engineering Body of Knowledge (LEBoK); the introduction to this volume. — Retrieved from: _____

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